What count telemetry can and cannot reveal about detector-control attacks on decoy-state QKD: a drift-aware, finite-key benchmark

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Abstract

Machine-learning intrusion detectors for quantum key distribution (QKD) report near-perfect accuracy on simulated attacks with mostly stationary benign data. We ask what the count telemetry of a three-intensity decoy-state BB84 link (per-window pulse, click, sifted-click and error counts with intensity-monitor readings) can reveal about detector-control attacks. The boundary is elementary: a faked-state attacker who reproduces the honest photon-number yields and error probabilities leaves the law of the counts unchanged. We benchmark thirteen detectors on simulated 24-hour streams with sifting, afterpulses, basis-dependent efficiencies and correlated slow drift at five scales, against three in-model and ten faked-state attacks, with paired honest copies and a ground-truth audit of key bits the attacker knows, on 40 seeds held out from development. Detectors that compare absolute levels with drift-free calibration alarmed in 74.57–97.27% of benign windows under nominal drift; drift-calibrated detectors, including a rolling-reference baseline, were over-conservative inside their calibration range and reached 30.71% at fourfold drift. A decoy-consistency statistic that refits the channel in every window stayed at 0.10–0.11% on average at all scales with drift-free calibration, given a per-window intensity monitor, and flagged naive blinding in the first 10-second window; its sequential forms caught part of the error-free and background-free emulation that the refit absorbs. No detector separated, at any operating point, an attacker who estimated the channel from five minutes of public counts, and a threshold-detector attacker was separated only by a post-hoc ten-minute block statistic. Except for an intermittent attack, every faked-state attack breached the finite-key bound in nearly every block that produced a key, with the attacker knowing over 98.97% of the sifted key. A 1% step in the intensity-monitor calibration raised false alarms sharply, so the monitor is a precondition, and with nominal intensities slow drift invalidated the single-photon bound in up to 1370 of 5 760 honest blocks.

Keywords: quantum key distribution, decoy states, detector blinding, intrusion detection, concept drift, finite-key security, anomaly detection

1 Introduction

Quantum key distribution (QKD) moves the security argument from computational hardness to physics. Its information-theoretic guarantee holds only inside a device model; once the hardware departs from that model, an eavesdropper can learn the key while every monitored statistic looks normal [1, 2]. Detector blinding is the standard example. Bright illumination turns single-photon avalanche detectors into classical threshold detectors, and a faked-state eavesdropper then dictates which detector clicks [3, 4]; a full-field implementation against a running link left the quantum bit error rate (QBER) unchanged and gave the attacker the entire key [5].

Operators therefore watch telemetry. A deployed decoy-state BB84 link [6–9] reports, per monitoring window, detection counts and error counts for every intensity and basis, and long field trials already feed some of these signals back into polarisation and phase stabilisation [10–12]. A recent strand of work trains machine-learning (ML) classifiers on such telemetry and reports near-perfect detection of intercept-resend, photon-number-splitting (PNS), Trojan-horse and blinding attacks [13–17]. These results come from simulators. In most of them the benign class is stationary or randomised sample by sample, the classes are balanced, and the attacker is not calibrated against the statistics the defender monitors; the adversarial formulation of [17], with domain randomisation and block-level decisions, is the main exception and the closest work (Section 2).

Two questions follow that the cited studies do not settle. The first concerns the attacker: which attacks are visible in window counts at all, given that a capable eavesdropper can shape the statistics she produces? The second concerns the defender: how many false alarms does a telemetry detector raise when the fibre drifts for a day, as installed fibre does [10, 11], rather than sitting at its calibration point? A detector that alarms every few minutes on a healthy link will be switched off; a detector that misses the one attack that breaks the security proof adds nothing that the decoy-state parameter estimation does not already provide.

This paper studies both questions for three-intensity decoy-state BB84 with the finite-key analysis of Lim et al.[18], restricted to *count telemetry*: per-window pulse, click, sifted-click and error counts with Alice's intensity-monitor readings. The boundary of what such telemetry can show is elementary. An attacker who reproduces the honest photon-number yields and error probabilities leaves the law of the counts unchanged (Proposition 1), and the perfect eavesdropper of [5], demonstrated on an entanglement-based system without decoy states, is its analogue. The contributions lie in what happens inside that boundary.

- A simulation benchmark with Bob's basis choice and sifting, per-basis misalignment, afterpulses, basis-dependent efficiencies, slow intensity drift and an intensity monitor, run over 24-hour streams at five drift scales. It covers three in-model attacks and ten faked-state attacks, from a naive one to one that emulates the honest channel exactly; one needs only an attenuated threshold detector, one estimates the channel from public counts, two ignore part of the drift, one stays just inside physically plausible parameters, and one is intermittent. All design choices were frozen before the reported runs, which use seeds not seen during development. Every attacked window is compared with the same window of an honest copy, and every post-processing block is audited against a ground-truth count of the key bits Eve knows.

- A comparison of thirteen detectors, including an adaptive-reference baseline, that separates two design choices: what a detector is calibrated on, and whether its features are invariant to drift. Features built on a per-window refit of the honest channel, the decoy-consistency statistic (DCS) and detectors trained on its residuals, keep the false-alarm rate near target at every drift scale with drift-free calibration only, under two stated preconditions on the link. Level-based detectors need drift-calibrated thresholds and then miss the target in both directions.

- A map of what count telemetry sees along the attacker ladder. Naive and intermittent blinding are caught at once. Error-free and background-free emulation is caught only by sequential or drift-calibrated detectors. Emulation with plausible parameters, emulation from public estimates and exact emulation pass every detector; threshold-detector emulation is separated only by a post-hoc block-level statistic.

- Two side results with practical weight. A quantitative illustration of a known requirement: with nominal intensities, slow intensity drift of a few per cent makes the single-photon bound of [18] fail in a sizeable share of honest blocks. And physical plausibility floors on the fitted parameters, an obvious remedy for the blind spots of the refit, fail on ordinary operational changes and are evaded by an attacker who stays above them.

The scope is deliberately narrow. All data are simulated, drift amplitudes are assumptions, attacks are pulse-wise independent, and the claims are about what count telemetry can establish. Richer observables such as timing and coincidence statistics [19, 20] and device-level countermeasures [21–23] are outside the detectors studied here and are discussed in Section 8.

Section 2 reviews decoy-state QKD and prior work on quantum hacking and its detection. Section 3 fixes the channel and threat models, and Section 4 states the indistinguishability boundary. Section 5 defines the DCS. Sections 6 and 7 give the experimental design and the results, and Section 8 discusses limitations.

2 Background and related work

2.1 Decoy-state BB84 and finite-key security

Weak coherent pulses carry a Poisson photon number, so multi-photon pulses leak information through photon-number splitting [24]. The decoy-state method [7–9] sends several intensities at random; since an eavesdropper sees only the photon number, the yields y_n and error rates e_n cannot depend on the intensity, and linear programming over the observed gains bounds the single-photon yield and its phase error. Practical versions use one signal, one weak decoy and one vacuum-like intensity [25]. Lim et al.[18] give closed-form finite-key bounds for this three-intensity protocol, using Hoeffding-type deviations for the decoy estimates and a sampling bound for the phase error. Hoeffding's inequality presumes independent trials, so these decoy bounds are rigorous for attacks that act independently on each pulse; coherent attacks call for martingale-type concentration bounds, for example Kato's inequality as used in [26]. Tighter analyses exist [27], but the closed form suits a benchmark that evaluates thousands of blocks, and the attacks studied here are pulse-wise independent. The bounds also assume exactly known

intensities. Intensity fluctuations have been treated rigorously, for random errors of known range [28], for imperfect sources in the finite-key regime [29], and with improved bounds for practical decoy-state systems [26]. We do not use these analyses; we substitute the block mean of a per-window intensity monitor into the bounds of [18], which is a heuristic, and check its effect against the true single-photon counts that the simulation records (Section 7).

2.2 Quantum hacking

Implementation attacks exploit the gap between device and model [1, 2, 30]. Time-shift attacks use detector efficiency mismatch [31]; Trojan-horse attacks read the modulator settings through back-reflected light [32, 33]; blinding attacks remove the quantum behaviour of the detectors altogether [3–5], including superconducting nanowire detectors [34]; attacks on the source change the mean photon number that the decoy analysis assumes [35, 36], and an attacker who can influence device calibration can open a detector side channel [37]. Countermeasures act on the hardware or on richer detection statistics: photocurrent monitoring [21], random detector efficiency [22], whose assumptions were later broken in a commercial system [38], variable attenuation combined with detection and error statistics [39], real-time monitoring of inter-detection times [19], background-noise cancellation for low-noise detectors [40], a test on two-detector coincidences that bounds Eve's information [20], best-practice criteria for self-differencing detectors [41], and measurement-device-independent QKD [23], which removes detector side channels by design. Several of these are statistical tests, but on observables that the window counts studied here do not contain.

2.3 Telemetry-based attack detection

Machine-learning detectors were first proposed for continuous-variable QKD, where the shot-noise and excess-noise estimates form natural features [13, 14]. For discrete-variable QKD, Al-Kuwari et al.[15] train hybrid quantum-classical LSTMs on 10 329 simulated decoy-state samples over eight balanced scenarios. Isha et al.[16] classify a balanced set of 24 000 simulated BB84 sessions with gradient-boosted trees and report a false-negative rate of 0.0198 against 0.8477 for an 11% QBER threshold; they also compare with a CUSUM detector, which reaches 78.99% accuracy but at a high false-positive rate at the chosen threshold. Mohamed and Al-Kuwari [17] go furthest: they pose detection as a game against an adaptive adversary with bounded hardware, use block-level telemetry that includes decoy-state residuals, timing moments and detector imbalance, apply domain randomisation, accumulate evidence across blocks with a CUSUM, and weight missed detections by the loss of finite-key secret fraction computed with three-intensity decoy estimators and entropy accumulation. Machine learning has also been used on the defender's side for parameter optimisation [42].

Against this background the present study has a narrower aim. It does not propose a better classifier. It differs from [17], the closest work, in four respects: decisions are made per 10-second window rather than per post-processing block; benign data are 24-hour streams in which loss, efficiency, background, misalignment and intensities drift slowly and in a correlated way, rather than training blocks whose parameters are randomised independently; every attacked window is compared with the same window of an honest copy of the channel; and each block's key is

audited against a ground-truth count of the bits Eve knows rather than against the bound. With these tools it asks what count telemetry can reveal at all when the attacker reproduces the monitored statistics, how often a detector alarms on a healthy drifting link, and whether the finite-key analysis would have released a key the attacker knows.

2.4 Drift and false alarms

Field links drift. Polarisation and fibre length change over tens of minutes, and long-running systems close feedback loops on the QBER and the count rates to compensate [10]; network trials have since run QKD layers in field and production settings, the SwissQuantum network for more than a year [12] and the Tokyo network as a field testbed [43], so a monitor must cope with environmental variation over days rather than minutes. In the stream-learning literature this is concept drift [44]: a model fitted to one regime degrades when the input distribution moves. For a monitoring system the visible symptom is the false-alarm rate. Sequential change detection, from Page's CUSUM onwards [45], treats false alarms as a design constraint rather than a by-product; that view drives the design of the statistic in Section 5.

3 Channel and threat model

3.1 Protocol and telemetry

Alice emits phase-randomised weak coherent pulses at rate R with intensity $k \in \mathcal{K} = \{\mu_1, \mu_2, \mu_3\}$, $\mu_1 > \mu_2 + \mu_3$, $\mu_2 > \mu_3 \geq 0$, chosen with probabilities p_k, in basis Z (key) with probability p_Z^A or X (test) otherwise. Bob measures in Z with probability p_Z^B, independently, with two threshold detectors, and announces which pulses produced a click and in which basis; a click is sifted when the two bases agree. Time is divided into monitoring windows of length Δ. Window t produces the *count telemetry*

$$\mathbf{T}_t = \left\{N_{a,k}(t),\, n^{\mathrm{all}}_{a,k}(t),\, n_{a,k}(t),\, m_{a,k}(t),\, \hat{\mu}_k(t)\right\}_{a\in\{Z,X\},\, k\in\mathcal{K}} : \tag{1}$$

the pulses Alice sent in basis a with intensity k, all clicks Bob announced for them, the sifted clicks, the sifted bit errors, and the per-window reading of Alice's intensity monitor. Timing, coincidence, afterpulse-correlation and photocurrent data are not part of $\mathbf{T}_t$; all statements below are about this count-level telemetry. A post-processing block aggregates B consecutive windows and runs the finite-key analysis of [18] with the block mean of the monitored intensities. It returns a key length ℓ (zero means abort).

3.2 Honest channel and benign drift

Conditioned on an n-photon pulse, the honest channel with transmittance η (fibre and detector), background yield Y_0 and misalignment e_d^b in basis b produces a click with probability, and a sifted error with probability,

$$y_n = 1 - (1 - Y_0)(1 - \eta)^n, \qquad e_{b,n}\, y_n = e_0 Y_0 + e_d^b\,(y_n - Y_0), \tag{2}$$

with $e_0 = 1/2$ [25]. The gain of intensity k is $Q_k = \sum_n P(n|k)\, y_n = 1-(1-Y_0)e^{-\eta k}$. Two further benign effects are included. Afterpulses add, in every gate, a click with probability $p_{\mathrm{ap}}\bar{Q}$, where $\bar{Q}$ is the mean gain over all pulses; such a click falls on an unrelated later pulse and so carries a random bit. Because $\bar{Q}$ is common to all pulses of a window, the afterpulse background is the same for the three intensities and enters the decoy analysis as part of Y_0; at the nominal values it makes up about 97% of the vacuum-like yield. Modelling it as an independent random-bit click per gate ignores its correlation with the preceding detection, which is a simplification of real detectors. Alice's intensities drift slowly, and her monitor reads them with a small error. Bob's two measurement paths and the polarisation-dependent loss of the link make the click probability depend slightly on Bob's and Alice's bases. In window t the parameters evolve as

$$L(t) = \alpha\ell_f + d\,[A_L \sin(\omega t + \phi) + U_L(t)]\ \ \mathrm{dB}, \qquad \eta(t) = \eta_B\,[1 + d\,U_B(t)]\,10^{-L(t)/10}, \tag{3}$$

$$p_{dc}(t),\, p_{\mathrm{ap}}(t) \propto 1 + d\,A_D \sin(\omega t + \phi - 0.5), \qquad e_d^b(t) = e_{\mathrm{mis}}^b + d\,s_j^b\,(t - t_j), \quad t \in [t_j, t_j + T_c), \tag{4}$$

$$\mu_k(t) = \mu_k\,[1 + d\,U_{I,k}(t)]\,(1 + \xi_{k,t}), \qquad \hat{\mu}_k(t) = \mu_k(t)\,[1 + \beta_k(t)]\,(1 + \zeta_{k,t}), \tag{5}$$

$$\Pr[\text{click},\, b \mid n, a](t) = p_b^B\, y_n(t)\,[1 + d\,U_{A,a}(t)]\,[1 + d\,U_{P,b}(t)], \tag{6}$$

where $\omega = 2\pi/24\,\mathrm{h}$; U_L, U_B, $U_{I,k}$, $U_{P,b}$ (efficiency of Bob's basis-b path) and $U_{A,a}$ (transmission of Alice's basis-a states) are Ornstein–Uhlenbeck processes, and β_k is a slow monitor bias, zero in the main runs and varied in Section 7.5; the misalignment of each basis grows at its own random slope s_j^b until an active compensation at period T_c resets it; $\xi_{k,t} \sim \mathcal{N}(0, \sigma_I^2)$ is fast per-window intensity noise and $\zeta_{k,t} \sim \mathcal{N}(0, \sigma_M^2)$ the monitor error. The scale d switches slow drift off ($d = 0$, the calibration regime) or amplifies it. Table 1 lists the values. They are assumptions chosen to reproduce the qualitative behaviour reported for installed fibre, slow loss variation and misalignment growth between compensations [10, 11]; they are not fitted to a specific link, and Section 7.5 varies the most influential ones. A mismatch between the two detectors of one basis, which would make the error rate depend on the bit value, and detector dead time are not modelled.

3.3 Threat model

Eve controls the quantum channel and may illuminate Bob's detectors. She cannot read Alice's intensity or basis choices before the public announcement, and she sees only what is announced in the classical post-processing. We restrict her to attacks that act independently and identically on every pulse given its photon number (collective attacks at the pulse level), so that each attack is described by the functions $(y_n, e_{b,n}, \kappa_n)$, where y_n is Bob's click probability, $e_{b,n}$ the sifted error probability in basis b and κ_n the fraction of sifted clicks whose key bit Eve knows exactly. In the simulator the basis factors $[1 + dU_{A,a}][1 + dU_{P,b}]$ multiply attacked and honest click probabilities alike. For the Alice-basis factor this places the polarisation-dependent loss before Eve, so that her detections inherit it; for the path factor of Bob it assumes that Eve emulates it, which an attacker who reproduces the honest law must do anyway and which is immaterial for the four-residual statistics, which do not use basis balance. Blind-static and Blind-track therefore still

Table 1: Simulation parameters. Device values follow the numerical example of [18]; drift amplitudes are scaled by d.

Quantity	Symbol	Value
Pulse rate, window length	R, Δ	100 MHz, 10 s
Intensities, probabilities	μ_k, p_k	(0.5, 0.1, 2×10^{-4}), (0.7, 0.2, 0.1)
Basis probabilities (Alice, Bob)	p_Z^A, p_Z^B	0.9, 0.9
Fibre length, attenuation	ℓ_f, α	50 km, 0.2 dB/km
Detector efficiency, dark count	η_B, p_{dc}	0.10, 6×10^{-7} per gate
Afterpulse probability	p_{ap}	1%
Base misalignment (Z, X)	e_{mis}^Z, e_{mis}^X	1.0%, 1.2%
Diurnal loss amplitude; OU loss s.d., time constant	A_L; σ_L, τ_L	0.3 dB; 0.15 dB, 30 min
Detector-efficiency OU s.d. (relative), time constant	σ_B, τ_B	2%, 60 min
Dark-count and afterpulse diurnal swing (relative)	A_D	10%
Misalignment growth per basis, compensation period	s_j^b, T_c	$U(0.5, 1.5)\times$1%/h, 1 h
Slow intensity drift: OU s.d. (relative), time constant	σ_{IU}, τ_I	1%, 2 h
Fast intensity noise; monitor error (relative)	σ_I; σ_M	1%; 0.2%
Monitor bias: OU s.d. (relative), time constant	σ_β, τ_β	0; 2 h
Bob basis-path efficiency: OU s.d. (relative), time constant	σ_P, τ_P	0.5%, 60 min
Alice-basis transmission (PDL): OU s.d. (relative), time constant	σ_A, τ_A	0.25%, 30 min
Post-processing block	B	60 windows (6×10^{10} pulses)
Security parameters	$\varepsilon_{\mathrm{sec}}$, $\varepsilon_{\mathrm{cor}}$, f_{EC}	10^{-10}, 10^{-15}, 1.16
Stream length		8 640 windows (24 h)

carry the basis factors. We also take Alice's source and intensity monitor as trusted: attacks that change the mean photon number or the monitor reading [35, 36] are outside this study. Coherent or adaptive attacks across pulses are outside this description. Table 2 lists the attacks; where the cited attack descriptions match Bob's signal gain, so do ours.

The attacks differ in what Eve knows. IR, PNS, Blind-naive, Blind-gain, Blind-nobg, Blind-thresh, Blind-cal and Blind-sub are given the current window's channel parameters $\eta(t)$, $Y_0(t)$ and $e_d^b(t)$; they are oracle attackers and bound what such knowledge buys. Blind-static knows only the nominal values and Blind-track additionally the fibre loss, which Eve can measure on the line herself. Blind-lag uses only public information: it estimates $(\eta, Y_0, e_d^Z, e_d^X)$ from the announced counts of the 30 windows (5 minutes) before onset with the refit of Section 5, then tracks the fibre loss and keeps the detector parameters frozen.

All faked-state variants use the same mechanism. Eve measures each pulse in Z or X with probability 1/2 and resends a bright faked state in her basis; a blinded detector pair clicks only when Bob's basis equals hers [3, 4]. To give Bob the same click probability y_n in both bases she forwards with probability $2y_n$, which requires $y_n \leq 1/2$. A click she injects on an empty pulse carries no information on Alice's bit, so it has error probability 1/2 and $\kappa_0 = 0$; on a non-empty pulse her measurement basis equals the sifted basis and $\kappa_n = 1$. Blinded detectors produce neither dark counts nor afterpulses unless Eve emulates them.

4 What count telemetry can and cannot reveal

Under the threat model the window counts are sums of independent indicator variables whose parameters depend on the attack only through $(y_n, e_{b,n})$. This gives the following statement, which is elementary and is not claimed as a contribution in itself; it fixes the boundary against which the benchmark is read.

Proposition 1 (Telemetry indistinguishability)**.** *Consider attacks that act independently and identically on pulses given their photon number. Let $y_n^{a,b}(t)$ be the honest probability that an n-photon pulse sent by Alice in basis a produces a click that Bob registers in basis b, and $e_{b,n}(t)$ the honest error probability of a sifted click in basis b, in window t; afterpulse clicks are included in y_0 and carry an independent random bit. Let an attack induce $\tilde{y}_n^{a,b}(t)$ and $\tilde{e}_{b,n}(t)$. If $\tilde{y}_n^{a,b}(t) = y_n^{a,b}(t)$ and $\tilde{e}_{b,n}(t) = e_{b,n}(t)$ for all n, a, b and t, the joint law of $(\mathbf{T}_1, \ldots, \mathbf{T}_W)$ under the attack equals the honest law. For every detector ϕ that maps count telemetry to alarms, $\Pr_{\text{attack}}[\phi = 1] = \Pr_{\text{honest}}[\phi = 1]$, the ROC curve of ϕ is the diagonal, and the finite-key length ℓ has the same distribution in both cases.*

Proof. Given Alice's settings (a, k) for a pulse, its photon number is Poisson with the realised mean $\mu_k(t)$ and independent of everything else; the monitor reading depends on Alice's source only. Whether the pulse produces a click registered in Bob's basis b, and whether a sifted click $(b = a)$ is in error, are then categorical variables, independent across pulses, with probabilities

$$\sum_n P(n|k)\, \tilde{y}_n^{a,b}, \qquad \frac{\sum_n P(n|k)\, \tilde{y}_n^{a,a} \tilde{e}_{a,n}}{\sum_n P(n|k)\, \tilde{y}_n^{a,a}}.$$

These coincide with the honest ones, so the counts of (1) have the same multinomial–binomial law in every window, and so does any function of them, such as ϕ and ℓ. □

The perfect eavesdropper of [5], on an entanglement-based link without decoy states, is an analogue. Independence across pulses is part of the hypothesis: clicks that are correlated between consecutive pulses, whether through the attack or through real afterpulsing, are not covered by the proof, and their effect on window counts would have to be bounded separately. The hypothesis is achievable with the mechanism above whenever $y_n \leq 1/2$, which holds here for every n with Poisson weight above 10^{-30}; if Eve chose Z with probability $q \neq 1/2$ and adapted her forwarding probability to her basis, the condition would become $y_n \leq \min(q, 1-q)$. The mechanism also assumes that the blinded detectors are controlled perfectly and produce no double clicks. Blind-cal meets it exactly. Blind-thresh shows that a photon-number-resolving detector is not needed in practice: with an attenuated threshold detector of efficiency $\eta_E \approx 2\eta$, the induced yields for $n \geq 2$ differ from the honest ones by $O(\eta^2)$ in absolute and $O(\eta)$ in relative terms, which changes the decoy gain by about 0.2% here. Both attacks require knowledge of $\eta(t)$, $Y_0(t)$ and $e_d^b(t)$, or of their effect on the counts; an eavesdropper who owns the fibre can estimate the loss from her own measurements of the line. Blind-static, Blind-track and Blind-lag model attackers with less knowledge. The proposition says nothing about richer telemetry. Coincidence statistics [20], inter-detection timing [19], variable attenuation at Bob [39] or background-noise cancellation [40] add observables whose law a faked-state attacker does not reproduce automatically, and these are where the hypothesis can fail.

Remark 1 (In-model attacks)**.** *Intercept-resend and PNS act only on the quantum channel, as does added loss. The finite-key bound of [18] is $\varepsilon_{\mathrm{sec}}$-secret against such attacks within its assumptions, which include a trusted source with known intensities, so every key it outputs is already secure; a telemetry detector can shorten the time to incident response or flag a denial of service, but it cannot add secrecy. For attacks outside the model the bound can certify a key that Eve knows. Telemetry detection matters for security only in the second case, and Proposition 1 limits it to attackers who do not reproduce the honest statistics.*

To audit secrecy in the simulations without trusting the bound, every sifted click carries a ground-truth flag stating whether Eve knows its key bit. Let Z be the sifted key of a block, C the public communication and E Eve's system, and let K_E of the n_Z bits of Z be known to Eve exactly. The key K is a function of Z and C, so for von Neumann entropies $H(K|EC) \le H(Z|EC) \le H(Z|E) \le n_Z - K_E$. If the key is ε-close in trace distance to a uniform key of length ℓ independent of EC, the Alicki–Fannes–Winter continuity bound gives $H(K|EC) \ge (1-\varepsilon)\ell - (1+\varepsilon)h\big(\varepsilon/(1+\varepsilon)\big)$. A block with

$$\ell > \frac{n_Z - K_E + (1+\varepsilon)\, h\big(\varepsilon/(1+\varepsilon)\big)}{1-\varepsilon}, \qquad \varepsilon = \varepsilon_{\mathrm{sec}} = 10^{-10}, \tag{7}$$

is inconsistent with $\varepsilon_{\mathrm{sec}}$-secrecy of that block's key. We call such a block a *breach*. Strictly, ε-secrecy is a statement about the ensemble of runs, weighted by the probability of not aborting; a breach frequency that is far above $\varepsilon_{\mathrm{sec}}$, as found here for the faked-state attacks, contradicts it. The test is one-sided: $\ell \le n_Z - K_E$ does not prove secrecy, because Eve may hold further partial information. It also has power only when Eve knows more than a fraction $1 - \ell/n_Z$ of the sifted bits, about 75% at the honest key rate, so it cannot confirm Remark 1 for intercept-resend or PNS; for those the security of the released key rests on the proof, not on the audit. For the faked-state attacks K_E follows from κ_n by construction, so the breach share measures how often the bound outputs a key at all under the attack; it is an audit of the accounting, not an independent finding. The simulation also records the true single-photon detections, which must exceed the lower bound $s_{Z,1}$ in honest operation.

5 Decoy-consistency statistic

Drift moves four physical parameters of the honest model: η, Y_0 (which also absorbs afterpulses, since they add a nearly constant background) and the misalignment in each basis. The count telemetry of one window has far more degrees of freedom. The decoy-consistency statistic (DCS) spends four of them on tracking the channel and tests the rest.

Per-window refit. Let $Q_k = \sum_a n^{\mathrm{all}}_{a,k} / \sum_a N_{a,k}$ be the gain of intensity k from all announced clicks, and use the monitored intensities $\hat{\mu}_k$. A fixed-point iteration on the signal and vacuum-like gains,

$$\hat{\eta} = -\frac{1}{\hat{\mu}_1} \ln \frac{1 - Q_{\mu_1}}{1 - \hat{Y}_0}, \qquad \hat{Y}_0 = \frac{Q_{\mu_3} - \big(1 - e^{-\hat{\eta}\hat{\mu}_3}\big)}{e^{-\hat{\eta}\hat{\mu}_3}}, \tag{8}$$

started from the linear approximation $\hat{Y}_0 = Q_{\mu_3} - (\hat{\mu}_3/\hat{\mu}_1)Q_{\mu_1}$ and run for six steps, gives $(\hat{\eta}, \hat{Y}_0)$. The sifted signal error rate E_b of each basis then gives $\hat{e}_d^b = (E_b Q_{\mu_1} - e_0\hat{Y}_0)/(Q_{\mu_1} - \hat{Y}_0)$, clipped to $[0, 1/2]$.

Residuals. With these estimates the honest model (2) predicts the observations that were not used in the fit, and four binomial residuals

$$r_j = \frac{c_j - M_j\hat{p}_j}{\sqrt{M_j\hat{p}_j(1-\hat{p}_j)}}, \qquad j = 1, \ldots, 4, \tag{9}$$

compare an observed count c_j out of M_j trials with its predicted probability $\hat{p}_j$. They are the decoy gain, the sifted decoy error rate in Z and in X, and the vacuum-like error rate (bases pooled, with the mean of $\hat{e}_d^Z$ and $\hat{e}_d^X$). The decoy-state estimator uses the same relations to bound the single-photon contribution over a block; the DCS turns them into a per-window test with a controlled false-alarm rate. The afterpulse background is taken as constant across the three intensities within a window, which holds to first order because the intensities are interleaved pulse by pulse.

Three further residuals are available in principle: the sifting fraction of signal clicks in each of Alice's bases, which equals p_a^B if Bob's click probability is independent of his basis choice, and the share of signal clicks on pulses Alice sent in Z, which equals the share of such pulses if the yield is independent of Alice's basis. They test the basis-independence hypothesis of Proposition 1 directly, but at 10^6–10^7 clicks per window they resolve relative changes of order 10^{-3}, below the slow drift of the efficiency of Bob's two measurement paths and of polarisation-dependent loss in deployed links. In our model with 0.5% path-efficiency and 0.25% polarisation-dependent-loss drift (Table 1) the seven-residual statistic (DCS-7) raised false alarms in 39.40% of windows at $d = 1$. We therefore report DCS-7 only as an ablation in Section 7.5. A link whose basis-path efficiencies are stabilised or monitored to 10^{-3} could add these residuals back.

Statistic and threshold. The statistic is

$$S_t = \sum_{j=1}^{4} \left(\frac{r_j - \mathrm{med}_j}{1.4826\,\mathrm{MAD}_j} \right)^2, \tag{10}$$

with the median and median absolute deviation taken from one benign stream recorded without slow drift. The robust standardisation absorbs effects that inflate a residual beyond its binomial variance, such as the fast intensity noise that the monitor does not remove. An alarm is raised when S_t exceeds the $(1-\alpha)$ quantile of S on separate drift-free benign calibration windows. A sequential version accumulates evidence across windows with Page's CUSUM [45]: $C_t = \max\{0, C_{t-1} + (S_t - \bar{S})/s_S - k\}$ with $k = 1/2$, where $\bar{S}$ and s_S are the calibration mean and standard deviation, and C_t is thresholded in the same way. A block version (DCS-block) averages the standardised residuals over the last $L = 60$ windows (10 minutes), $\bar{z}_{j,t} = L^{-1}\sum_{s=t-L+1}^{t}(r_{j,s} - \mathrm{med}_j)/(1.4826\,\mathrm{MAD}_j)$, standardises each $\bar{z}_j$ again with the median and MAD of the same block means on the calibration stream (which absorbs their autocorrelation), and sums the squares; it targets small persistent shifts that a single window cannot resolve. We added it after the

first round of review, in response to a small but systematic shift of the Blind-thresh decoy gain (Section 7.3); it is therefore a post-hoc detector, and its thresholds and window length were fixed before the final runs.

Disclosure, cost and blind spots. The DCS needs, per window, the basis and intensity of every pulse with its click outcome, which Alice and Bob reveal anyway during sifting and parameter estimation, and the error counts per basis and intensity. Errors in the test basis X are compared publicly and are available in real time. Errors in the key basis Z are known only after error correction; the DCS assumes that error correction runs on frames of one window (about 10^6 sifted bits here, a common frame size), so its decisions lag the window by the error-correction latency, and Blind-lag is assumed to read the same per-window error rates from the public discussion. Where Z errors are not available per window, the variant DCS-X fits only $\hat{e}_d^X$ and uses three residuals (decoy gain, decoy X error rate and vacuum-like X error rate); it is evaluated alongside the DCS. Scoring a single window costs about 0.1 ms in our Python implementation (vectorised over a day, 0.7 μs per window). Slow drift of η, Y_0, e_d^Z or e_d^X moves the estimates, not the residuals. The refit also defines what the DCS cannot see. Any attack whose window statistics coincide with an honest channel at *some* parameter values passes unseen, whether or not those values are physically plausible: pure added loss looks like lower transmittance, a missing background looks like a lower Y_0, and error-free faked clicks look like zero misalignment. Section 7.5 tests whether plausibility floors on the fitted parameters (DCS+F) can close this gap; they cannot without false alarms on ordinary operational changes.

6 Experimental design

6.1 Streams and protocol

Each realisation (seed) generates its own channel trajectories, calibration data, training data and test streams; seeds are the unit of replication. A test stream covers 24 hours (8 640 windows). Benign streams are run at drift scales $d \in \{0, 0.25, 1, 2, 4\}$; $d = 0.25$ represents a well-stabilised link with misalignment growth of about 0.25%/h. For each scale and each attack of Table 2, a further stream carries the attack for two hours (720 windows) starting at a uniformly random window between hour 4 and hour 20. The same channel trajectory is also sampled without the attack, which gives a paired honest copy of the attacked windows. Counts are drawn per photon-number component ($n = 0$, $n = 1$, $n \geq 2$) and then thinned by Bob's basis choice, so every block records its true single-photon detections and the number of sifted key bits Eve knows. The finite-key analysis runs on every complete block of 60 windows, on a fixed grid of the stream, that lies inside the attack, and on all 144 blocks of each benign stream.

6.2 Detectors

Thirteen detectors share one interface: a score per window and a threshold.

- **QBER**: the sifted Z-basis QBER of the signal intensity against the fixed 11% limit used as the baseline in [16].

- **Band**: a two-sided physics band. The signal QBER and the decoy-to-signal gain ratio per unit monitored intensity are standardised with the median and MAD of a drift-free benign stream, and the score is the larger absolute z-value. It alarms on QBER that is too low as well as too high.

- **DCS**, **CUSUM** and **Block** (DCS-block): Section 5. CUSUM and Block are sequential: their window scores are not independent, so their per-window FPR is not directly comparable with that of the single-window detectors, and alarm episodes and time to first false alarm are the more relevant measures for them. The CUSUM does not reset after an alarm, so it stays in alarm while the evidence persists; its per-window TPR measures how much of the attack it covers, not how fast it detects. A CUSUM on the Band features would inherit Band's drift failure (Section 7.2) and is not included. **DCS-X** is the variant that uses only test-basis errors (Section 5).

- **Band-roll**: the Band features with an adaptive reference, each minus its causal rolling median over the previous hour (360 windows), standardised and thresholded on drifting benign data. It represents the rolling baselines operators use; a shorter reference absorbs a two-hour attack after onset.

- **IF-stat**, **IF-drift** and **IF-res**: isolation forests [46] with 200 trees. IF-stat and IF-drift use nine features: the logarithm of the gain per intensity and the sifted QBER per basis and intensity. IF-stat is fitted on one drift-free benign stream, IF-drift on three benign streams with $d \in \{0.5, 1, 2\}$. IF-res is fitted on the four standardised DCS residuals of the drift-free stream.

- **RF-stat**, **RF-DR** and **RF-res**: random forests [47] (200 trees, minimum leaf size 2) trained as binary attack-versus-benign classifiers in the style of the literature [15, 16]. The attack class holds six streams of 2 160 windows (IR with $f \in \{0.05, 0.1, 0.25, 0.5\}$, PNS, Blind-naive). Added loss is a benign event from the protocol's point of view and is not used as an attack label. RF-stat sees one drift-free benign stream and no drift. RF-DR sees six benign streams and draws a drift scale from $U(0, 4)$ for every training stream, the domain randomisation of [17]. RF-res uses the same data as RF-DR with the standardised DCS residuals as features. IR at $f = 0.05$, PNS and Blind-naive are therefore in the training distribution of the supervised detectors (marked with * in the tables); no detector sees the other blinding variants.

6.3 Thresholds

Every threshold except the fixed QBER limit is the $(1 - \alpha)$ quantile, $\alpha = 10^{-3}$, of the detector's scores on three calibration streams (25 920 windows), with randomised tie-breaking: a score equal to the threshold alarms with the probability that makes the calibration alarm rate exactly α. Without it, a random forest whose calibration scores pile up at 0 or 1 would either never or always alarm. Two calibration sets exist. The drift-free set is what an operator has after commissioning; it is the default for every detector not built to use drift. IF-drift, RF-DR, RF-res and Band-roll are drift-aware by design and default to the drifting set ($d \in \{0.5, 1, 2\}$, marked

†). Each detector is also evaluated with the other set as a control. Test drift $d = 4$ lies outside both calibration sets; for RF-DR and RF-res it lies at the edge of the training range.

Seed history. The benchmark was developed on seeds 0–9, which were run several times while the design changed in response to review: four instead of seven residuals, the Block, DCS-X and Band-roll detectors, the basis-dependent drift terms, the monitor bias, the attackers Blind-lag, Blind-sub and Blind-partial, and the block-wise release policy. The design was then frozen, and every result reported here comes from seeds not used during development: seeds 10–49 for the main benchmark, 500–509 for DCS+F, 600–609 for the stress test and 700–709 for the sensitivity runs. DCS+P, the first floor variant, failed during development and is reported from its development seeds 200–209. The Block detector and the floor variants remain post-hoc in the sense that they were motivated by development results. IR at $f = 0.2$ lies between two training strengths of the supervised detectors and is marked as interpolated (‡).

6.4 Metrics and statistics

For each detector the benchmark records:

- the per-window false-alarm rate on benign streams (FPR, expected alarm probability under the randomised threshold) and the number of alarm episodes per day, an episode being a maximal run of consecutive alarmed windows;
- the true-positive rate on attacked windows (TPR);
- the paired ROC AUC of the 720 attacked windows against the same 720 windows of the honest copy, which removes time-of-day confounding between attacked and non-attacked parts of a stream. It is the Mann–Whitney statistic of two time-matched samples and does not exploit the pairing further;
- the delay from attack onset to the first alarm, in windows;
- the TPR at a matched FPR, with the threshold set at the $(1 - \alpha)$ quantile of the detector's own benign scores at the same drift scale. This oracle threshold is estimated in-sample from about nine exceedances, is not available to an operator, and is one-sided, so it cannot register attacks that make windows look more normal; it separates a detector's discriminating power from its calibration;
- the time to the first false alarm on benign streams, an empirical in-control run length, which is the more relevant measure for the sequential detectors.

An AUC below 0.5 means the attack makes windows look *more* normal than honest operation. A one-sided detector does not alarm on such windows, but the attack is still separable, and we report it as such. For each block the benchmark records ℓ, n_Z, K_E, the breach indicator (7), the bound checks $s_{Z,1} \leq$ true single-photon count in Z and $s_{X,1} \leq$ its X counterpart, and the phase-error check $v_{X,1} \geq$ true single-photon errors in X. It also records the same quantities with nominal instead of monitored intensities, and runs the same analysis on the paired honest copy. To show what an alarm buys, we apply a block-wise release policy: a block is withheld by a detector if that detector alarmed in any of its windows, and release resumes with the next block.

Applied to the paired honest copy, the same policy gives the cost in withheld honest blocks, so that withholding by detection can be told apart from withholding by false alarms.

Means over the 40 seeds carry 95% BCa bootstrap intervals over seeds [48] (10^4 resamples, a fixed generator per interval). When all seeds give the same value the interval is reported as degenerate, and when the BCa correction is undefined the percentile interval is used instead. Because a signed AUC averaged over seeds can hide separations of opposite sign, we also report the mean of $|\mathrm{AUC} - 0.5|$ per seed for the attacks claimed to be invisible. Differences between the DCS and another detector are paired by seed and bootstrapped the same way. We report effect sizes with intervals rather than significance tests. Since one seed corresponds to one link, we also give the largest per-seed false-alarm rate and the share of seeds above 2α. One seed of the full benchmark takes about 127 s on four CPU cores.

7 Results

7.1 Honest operation and the bound check

The simulator reproduces the expected finite-key behaviour, and a unit test confirms that with expected counts and a very large block the finite-key length approaches the asymptotic rate computed from the true single-photon yield, from below. At the calibration point the honest link yields 9.38×10^{-4} secret bits per pulse at a sifted QBER of 1.50%. Nominal drift lowers the rate to 7.66×10^{-4} and fourfold drift to 3.77×10^{-4}, where 79.3% of the blocks still produce a key. At $d = 1$ the loss spans 9.1–10.8 dB and the misalignment reaches 2.7%; at $d = 4$ the ranges are 6.5–13.1 dB and 7.2%.

With the block-mean monitored intensities, the lower bounds $s_{Z,1}$ and $s_{X,1}$ never exceeded the true single-photon counts and the phase-error bound $v_{X,1}$ never fell below the true single-photon errors, in any of the 28800 honest blocks, and no honest block met the breach condition (7). With the nominal intensities, which is what the bounds of [18] assume, $s_{Z,1}$ exceeded the true count in 7, 239 and 1370 of the 5 760 blocks at $d = 1$, 2 and 4 (between 0 and 71 of 144 blocks per seed at $d = 4$), and never at $d \leq 0.25$. Since the substituted intensities are the only difference, slow intensity drift of one to a few per cent suffices to invalidate the single-photon bound in a sizeable share of blocks, where failures at a rate of order $\varepsilon_{\mathrm{sec}}$ would be expected. This quantifies a known requirement [26, 28, 29] rather than revealing a new one; the agreement with monitored intensities is an empirical check in this model, not a guarantee. The monitor is therefore part of the setting studied here, not an optional extra, and Section 7.5 shows that the DCS depends on it as well.

7.2 False alarms under drift

Table 3 and Figure 1 give the per-window false-alarm rate on benign streams. With drift-free calibration, every detector except the fixed QBER rule meets its 10^{-3} target at $d = 0$. Drift then splits them into three groups.

Detectors that compare absolute levels with a drift-free reference fail at once. At $d = 0.25$ the two-sided Band, IF-stat and RF-stat already alarm in 89.31%, 41.52% and 6.34% of benign windows, and at $d = 1$ in 97.27%, 95.05% and 74.57%. Their alarms follow the hourly

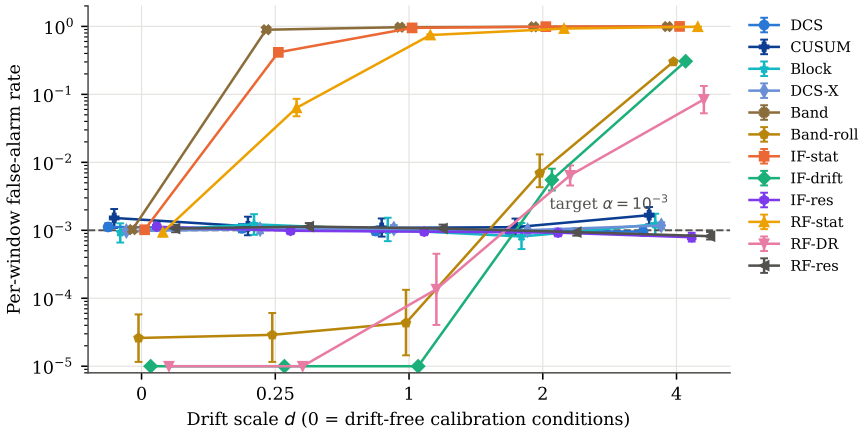


Figure 1: Per-window false-alarm rate against drift scale (log axis; markers are offset horizontally for legibility; error bars are 95% bootstrap intervals over 40 seeds; zero rates are drawn at 10^{-5}). The QBER rule raised no alarms and is not shown. The dashed line is the calibration target.

misalignment ramp (Figure 2, lower panels), which gives an operator 230.2 (IF-stat) and 472.1 (RF-stat) separate alarm episodes per day on a healthy link (Table 12).

Drift-calibrated detectors do not hold their target either. IF-drift, RF-DR and Band-roll, thresholded on benign data with $d \in \{0.5, 1, 2\}$, raise almost no alarm at $d \leq 1$ (at $d = 1$: 0.00%, 0.01% and 0.00%), which means they operate far below the target and give up detection power; at $d = 2$, the edge of the calibration range, they reach 0.55%, 0.64% and 0.69%, and at $d = 4$ 30.71%, 8.43% and 30.20%. RF-DR was trained on drift up to $d = 4$; its threshold was not, and with a drift-free threshold its false-alarm rate at $d = 4$ is 48.56% (Table 9).

Detectors built on the per-window refit stay near the target at every scale on average: the DCS between 0.10% and 0.11%, IF-res, RF-res and DCS-X between 0.08% and 0.12%, and the sequential CUSUM and Block between 0.08% and 0.17% (Table 3). For a single link, which corresponds to one seed, the spread matters: the largest per-seed DCS rate at $d = 4$ is 0.19% and 0% of seeds exceed 2α, against 0.59% and 35% for the CUSUM and 0.54% and 18% for the Block detector. The sequential detectors raise fewer alarm episodes (3.5 and 3.2 per day at $d = 1$, against 8.4 for the DCS) but in longer runs, and their first false alarm comes later (median 5.6 and 5.9 hours at $d = 1$, against 2.5 hours; no false alarm in a whole day in 9 and 14 of 40 seeds). At $d = 4$ the paired difference in false-alarm rate between the DCS and IF-drift is -30.61 percentage points (95% interval $[-32.95, -28.33]$); between the DCS and RF-res it is indistinguishable from zero ($[-0.00, 0.03]$). At $\alpha = 10^{-4}$ the expected count falls to 0.7 alarms per day at $d = 1$, while the TPR on Blind-naive stays at 1.00 and the TPR on IR with $f = 0.05$ falls to 0.51. The appendix gives all three levels of α (Tables 10 and 11).

7.3 Which attacks are separable

Figure 3 gives the paired AUC at $d = 1$ with its bootstrap half-width, and Table 4 the TPR at the calibrated threshold together with the benign false-alarm rate of the same detector. The appendix gives the AUC at $d = 0$, 0.25, 2 and 4 and the TPR at matched false-alarm rate.

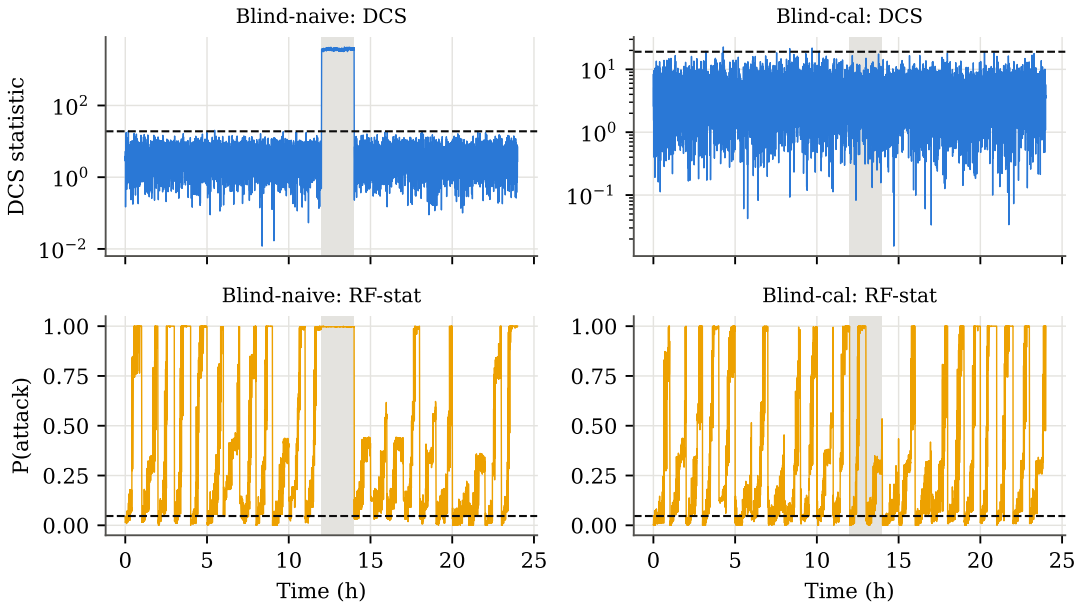


Figure 2: Window scores over one 24-hour stream at $d = 1$ with the attack active in the shaded two hours; dashed lines are the calibrated thresholds. Left: Blind-naive. Right: Blind-cal. Top: DCS (log axis). Bottom: RF-stat, whose score follows the hourly misalignment ramp and crosses its threshold in most benign windows.

Calibrated faked states are invisible. For Blind-cal the paired AUC of every detector at every drift scale lies between 0.49 and 0.51; the per-seed $|\mathrm{AUC} - 0.5|$ is at most 0.08 on average (the Block detector, whose ten-minute averages leave fewer effective samples), and the TPR at matched false-alarm rate is at most 0.01. Proposition 1 holds by construction for this attack, so this only validates the simulator. Blind-lag, which estimates the channel from public counts before onset, is invisible to every refit-based detector (per-seed $|\mathrm{AUC} - 0.5|$ of 0.01 for the DCS at $d = 1$, as for Blind-cal). Level-based detectors see per-seed differences of either sign, because a stale estimate drifts above or below the current channel (per-seed $|\mathrm{AUC} - 0.5|$ of 0.25 for RF-stat at $d = 1$), but these cancel on average (mean AUC 0.18–0.58) and no detector turns them into alarms: the TPR at matched false-alarm rate is at most 0.01 for every detector at every drift scale. Blind-thresh, which needs only an attenuated threshold detector, is not at chance: its 0.2% excess decoy gain shifts the single-window statistics slightly (AUC up to 0.58 without the Block detector; 41 of 65 intervals exclude 0.5), but no single-window detector that holds its false-alarm rate reaches a TPR above 0.01. The post-hoc Block detector, which averages the residuals over ten minutes, separates it with AUC 0.96 at $d = 1$ and covers 0.48 of the attacked windows (0.47 at $d = 4$). The high TPR of Band, IF-stat and RF-stat on these attacks in Table 4 equals their benign false-alarm rate and is not detection.

Imperfect emulation is visible, but not to every detector. At $d \leq 1$, Blind-naive is separated by every detector except Band-roll, whose rolling reference partly absorbs it (AUC 0.61), and the one-sided QBER rule, which scores AUC 0.00 because the attack lowers the QBER; the DCS raised its first alarm in the first attacked window in 200/200 runs. Blind-partial, which blinds a tenth of the pulses, is caught by the DCS with TPR 0.95. Blind-gain and

Blind-nobg separate the detectors. The single-window DCS refit absorbs both: error-free faked clicks look like zero misalignment, missing background clicks look like a lower Y_0, and the AUC is only 0.64 and 0.59 at $d = 1$. DCS-X, which sees no Z errors, misses Blind-gain entirely (AUC 0.48). The sequential detectors accumulate the small shift that remains: the CUSUM detected Blind-gain in 200/200 runs after a median of 21.5 windows and the Block detector covers 0.89 of the Blind-nobg windows; because the CUSUM does not reset, its per-window TPR (0.94 on Blind-gain, 0.34 on Blind-nobg) measures coverage rather than detection speed. RF-DR, which saw Blind-naive in training, generalises to both inside its calibrated range (TPR 0.93 and 0.48 at $d = 1$), and IF-drift separates both at matched false-alarm rate (TPR 1.00 and 1.00) but not at its own threshold. Blind-sub, which lowers background and misalignment only to plausible levels, passes every refit-based detector (AUC 0.50 for the DCS); IF-drift separates it at matched false-alarm rate (AUC 1.00) but reaches TPR 0.03 at its calibrated threshold. The rolling-reference Band-roll separates none of the faked-state attacks well (AUC at most 0.65 at $d = 1$).

Attackers who ignore drift differ from the honest copy in reverse. Blind-static emits the statistics of the nominal channel, and Blind-track tracks the fibre loss while replaying nominal detector values. Both remove part of the benign variation, so level-based detectors score the attacked windows as *more* normal than the paired honest windows: AUC 0.01 for the QBER and 0.00 for IF-stat on Blind-static, and 0.19 for RF-stat on Blind-track. This shows that their window statistics differ from the contemporaneous honest law. It does not show that an operator could detect them: the comparison uses the honest copy, which an operator does not have, and no detector in the benchmark alarms on missing variation. The DCS removes the difference altogether (AUC 0.50 and 0.50), because its refit absorbs the parameter levels that carry it.

In-model attacks. IR and PNS give DCS AUC of at least 0.96 at every drift scale, and the CUSUM detects IR with $f = 0.05$ in 200/200 runs with a median delay of 1 window. PNS lowers the QBER, so the one-sided QBER rule scores AUC 0.07 at $d = 0$. The refit-based detectors cannot see added loss (DCS AUC 0.47 at 3 dB), by construction.

7.4 Finite-key audit

Table 5 links detection to secrecy at $d = 1$. The finite-key analysis aborts every PNS block and every IR block at $f = 0.2$. At $f = 0.05$ it outputs a shorter key (3.87×10^{-4} per pulse, -50.2% relative to the paired honest copy) while Eve knows 2.50% of the sifted bits, and no block breaches. Added loss only shortens the key. No in-model attack produced a breach in any block at any drift scale (maximum 0.0%), and neither did Blind-partial, under which Eve knows 10.00% of the sifted bits; as noted in Section 4, the audit has no power at these fractions.

Every other faked-state variant did the opposite. Eve knew between 98.97% and 100.00% of the sifted key bits (the remainder are clicks on empty pulses and emulated afterpulses), and at least 96.0% of the blocks that produced a key breached at every drift scale. As noted in Section 4, this confirms the accounting rather than reporting a discovery: once the bound releases a key under a faked-state attack, that key is known. Blind-naive roughly triples the key length relative

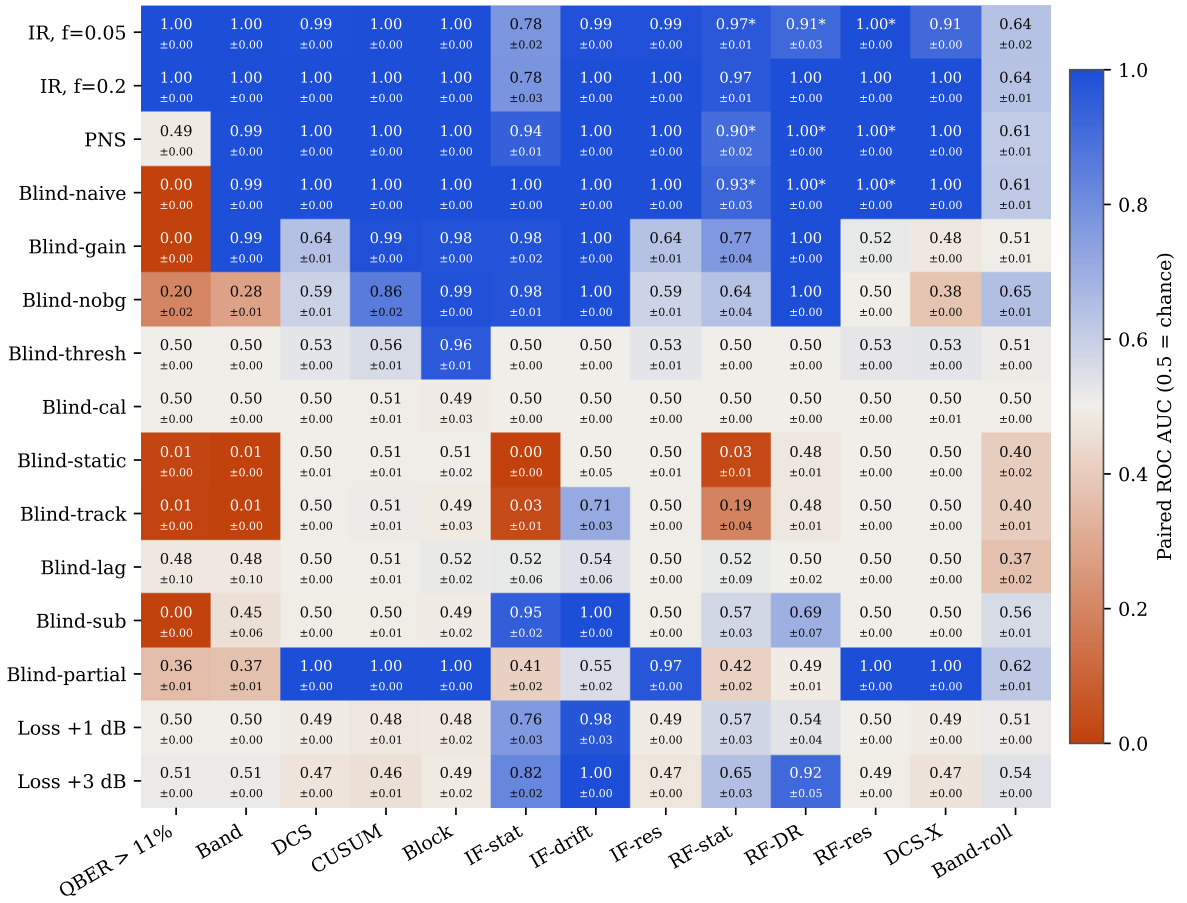


Figure 3: Paired ROC AUC at $d = 1$ (attacked windows against the same windows of the honest copy; mean over 40 seeds, with the half-width of the 95% bootstrap interval below each value). Blue: attacked windows score higher; orange: lower, meaning the attack makes windows look more normal; light: chance. *: attack in the training distribution of the supervised detector. IR with $f = 0.2$ lies between two training strengths.

to the honest copy (204.3%), because its faked clicks carry no errors.

Table 6 shows what the detectors buy under the block-wise release policy, together with its cost on the paired honest copy. Every breached Blind-naive block is withheld by the DCS, CUSUM, Block and DCS-X detectors, at a cost of 6% (DCS) and 2% (CUSUM) of honest blocks; per seed, the DCS withholds 94% more of the breached blocks than of the honest ones (95% interval 91–96). On Blind-gain the CUSUM and Block detectors withhold almost every breached block (excess 99% and 98%), the DCS 78%, and DCS-X only 17%. On Blind-nobg the Block detector withholds 98% in excess of its cost, the CUSUM 66%, and on Blind-thresh the Block detector releases 77 of 441 breached blocks. Against Blind-cal, Blind-lag, Blind-sub, Blind-static and Blind-track every refit-based detector releases more than 90% of the breached blocks, and the blocks it withholds are withheld at the same rate as honest blocks, by false alarms.

7.5 Sensitivity, preconditions and post-hoc variants

Table 8 varies one factor at a time on fresh seeds (ten per setting, 24-hour streams, two-hour attacks; the Block detector always averages ten minutes). The single-window DCS stays between

0.07% and 0.21% in every setting but one. Without the intensity monitor it rises to 1.11% at $d = 1$ and 31.14% at $d = 4$, because slow drift of the intensity ratios then enters the decoy-gain residual. The sequential detectors are more fragile. A slow monitor bias of 0.2%, 0.5% and 1% (relative s.d., not scaled by d) leaves the DCS at or below 0.21% but raises the CUSUM to 7.78%, 13.45% and 13.53% at $d = 4$; at 1% bias the single-photon bound with monitored intensities also failed in 3 of 1 440 blocks at $d = 1$. A larger but unbiased monitor error ($\sigma_M = 0.5\%$ and 1%) keeps every false-alarm rate near target but costs power (IR TPR 0.39 and 0.17 instead of 0.69; Block TPR on Blind-thresh 0.05 and 0.00). Without afterpulses the CUSUM and Block rates at $d = 4$ rise to 1.65% and 0.45%. The ablation DCS-7, which adds the three basis-balance residuals, raised false alarms in 39.40% of windows at $d = 1$ in the nominal setting, 6.80% with constant path efficiencies and 30.14% without polarisation-dependent loss, while the four-residual DCS was unaffected by either. A basis asymmetry in misalignment of up to one percentage point has no effect, since each basis has its own fitted $\hat{e}_d^b$.

Detection power depends on counts per window and per block. IR at $f = 0.05$ is caught by the DCS with TPR 1.00 in 60-second windows and 0.02 in 1-second windows, 1.00 at 25 km and 0.16 at 75 km, and an afterpulse probability of 5% lowers it to 0.22. Blind-naive is caught in every setting. Blind-gain and Blind-nobg pass the single-window DCS in every setting, including $p_{\mathrm{ap}} = 0$, where the background consists of dark counts only. The Block TPR on Blind-thresh is 0.51 in the nominal setting, 0.98 at 25 km and 0.01 at 75 km; with 1-second windows, where ten minutes hold 600 windows, it is 0.91, and with 60-second windows, ten, 0.05. With nominal intensities, the single-photon bound failed in 5 of 1440 honest blocks at $d = 1$ in the nominal setting and in 113 with doubled intensity drift; with monitored and unbiased intensities it never failed.

Benign operational events. Table 7 applies events outside the drift model to the second half of a 24-hour stream at $d = 1$ (fresh seeds 600–609). The refit-based detectors absorb a fivefold change in afterpulse probability or dark counts, a better misalignment compensator, a 7 dB loss step, a 20% efficiency drop and a monitored 3% laser re-lock. A single 1% calibration step of the decoy-intensity monitor does not leave them intact: the DCS alarms in 6.5% of windows, DCS-X in 7.3%, and the CUSUM and Block detectors stay in alarm for the rest of the day.

The blind spots of the refit suggested adding prior knowledge on the fitted parameters, which the refit otherwise treats as free. DCS+P (development seeds 200–209) adds one-sided robust z-floors on $\log \hat{Y}_0$ and $\hat{e}_d^b$; its false-alarm rate grows to 8.79%, 24.69% and 44.91% at $d = 1$, 2 and 4, because benign drift moves $\hat{Y}_0$ far beyond its drift-free spread. DCS+F (fresh seeds 500–509) alarms outright if $\hat{e}_d^b$ falls below half, or $\hat{Y}_0$ below a quarter, of its drift-free calibration median. On the main drift model it catches Blind-gain and Blind-nobg (AUC 1.00 and 1.00 at every drift scale) at the DCS false-alarm rate (0.09% at $d = 1$, 0.10% at $d = 4$), and at $d = 4$ the benign fitted parameters stay at least 1.80 times above the floors. It does not survive ordinary operational changes outside that drift model: a fivefold drop in afterpulse probability after a detector change, a better misalignment compensator and a 7 dB loss step, which lowers the gain-dependent afterpulse background, raise its false-alarm rate to 48.4%, 19.6% and 47.7% (Table 7). And Blind-sub, which lowers background and misalignment only to just above the floors, passes it (AUC 0.50 at $d = 1$, 0.59 at $d = 4$, where drift occasionally pushes the attacker's

parameters below the floors). We therefore report DCS+F as a negative result: fixed floors trade a blind spot for a false-alarm mode and move the blind spot rather than close it.

8 Discussion

8.1 What a telemetry detector is for

Faked-state attacks that reproduce most of the honest statistics breached almost every block that produced a key. Attacks inside the device model and the intermittent Blind-partial did not, but the audit cannot detect a breach unless Eve knows most of the sifted key, so for them it is consistent with Remark 1 rather than evidence for it; the decoy-state bound shortened or aborted their keys, and their security rests on the proof. A telemetry detector therefore adds secrecy only against model-violating attacks that would otherwise produce a key, and Proposition 1 limits even that role to attackers who leave a trace in the counts. Detection accuracy on intercept-resend or PNS, the headline numbers of several ML studies [15, 16], measures something the protocol already provides. Its operational value is different and real: faster incident response, and a signal that separates a denial of service from a degrading fibre. The game-theoretic formulation of [17], which weights missed detections by lost secret fraction computed from the bounds, points in this direction; the audit here adds a ground-truth check of whether a released key was in fact known, for the attacks where that check has power.

8.2 Drift, calibration and the choice of features

Two design choices decide whether a detector survives drift. The first is what it is calibrated on. Detectors with a drift-free threshold that compare absolute levels (Band, IF-stat, RF-stat) fail at the first sign of drift; at $d = 0.25$ the Band already alarms in 89.31% of windows. Detectors calibrated on drifting data (IF-drift, RF-DR, Band-roll) do not hold their target either: inside the calibrated range they are over-conservative (IF-drift 0.00%, RF-DR 0.01% at $d = 1$), and beyond it they exceed the target (30.71% and 8.43% at $d = 4$). Calibrating on a mixture of drift scales fixes a threshold for the mixture, not for any one link, and an adaptive reference does not remove the problem, because it must be slow enough not to absorb the attack. At a matched false-alarm rate, which needs an oracle threshold, IF-drift is strong at $d = 1$ (TPR 1.00 on Blind-gain and 1.00 on Blind-sub; Table 18), so inside its calibration range its weakness is calibration, not discrimination; at $d = 4$ it also loses discrimination (0.00 and 0.00; Table 19). The second choice is what the detector sees. Features that are invariant to drift by construction, the refit residuals, keep the false-alarm rate near target for every detector built on them, unsupervised or supervised, at every drift scale tested, with drift-free calibration only.

The same invariance defines the blind spots. Whatever the refit absorbs, a single-window residual detector cannot see: pure loss, the error-free clicks of Blind-gain, the missing background of Blind-nobg, and a background and misalignment that are lowered but still plausible (Blind-sub). Sequential use of the residuals recovers part of this: the CUSUM catches Blind-gain and the Block detector Blind-nobg, because a lowered fit is not exactly an honest fit and the small remaining shift accumulates. Plausibility floors on the fitted parameters do not help. Floors set from the drift-free spread (DCS+P) were crossed by ordinary drift, and fixed floors (DCS+F)

were crossed by ordinary operational changes and evaded by an attacker who stays just above them (Section 7.5). Any floor encodes an assumption about how far a healthy detector can move, and the attacker can read the same assumption.

8.3 The limits of the indistinguishability argument

Proposition 1 is an existence statement about pulse-wise attackers and count telemetry. The benchmark shows that its hypothesis is easier to meet than an oracle attacker suggests. Blind-thresh needs no photon-number-resolving detector, and only the post-hoc Block detector separated its 0.2% excess decoy gain, covering 0.48 of the attacked windows and withholding 81% more of the breached blocks than of honest ones at $d = 1$; ten minutes of averaging is what this attacker costs the defender. Blind-lag needs no access to the current channel parameters: five minutes of public counts before onset, plus the fibre loss that Eve can measure herself, were enough to pass every detector at every drift scale (AUC at most 0.58) and to breach 100.0% of blocks at $d = 1$. Its estimate goes stale as the detector parameters drift, but slowly enough that the refit absorbs it.

Attackers who replay nominal detector values (Blind-static, Blind-track) remove benign variation, and level-based detectors score their windows as more normal than the honest copy. This is a difference from the contemporaneous honest law, measured with an AUC that needs the honest copy. An operator has no such copy, and a rule that alarms on missing variation would also alarm on genuinely quiet periods; we did not test such a rule, so we make no claim that it would work. The DCS cannot use the difference at all, because its refit absorbs the levels that carry it.

The proposition also marks where telemetry can do better. Observables outside $\mathbf{T}_t$ whose law a faked-state attacker does not reproduce automatically break its hypothesis: two-detector coincidences [20], inter-detection timing [19], the response to random attenuation at Bob [39], or background-noise statistics of low-noise detectors [40]. Combined with hardware criteria [41], photocurrent monitoring [21] or measurement-device-independent protocols [23], these change the observation model itself. Count-level monitoring should be judged by its false-alarm rate on a drifting link and by the model-violating attacks it catches, with an explicit statement of the attacker it cannot catch.

8.4 Preconditions for the DCS

The DCS relies on properties of the link that the benchmark assumes and an operator must check. First, an intensity monitor must be present and stable. Without a monitor the bound itself fails and the DCS false-alarm rate rises to tens of per cent at $d = 4$; a slow monitor bias of a few tenths of a per cent multiplies the CUSUM false-alarm rate by up to two orders of magnitude while leaving the single-window DCS near target; and a single 1% calibration step of the decoy monitor raises the DCS false-alarm rate to 6.5% and drives the CUSUM and Block detectors into permanent alarm (Section 7.5). The DCS cannot tell such a monitor fault from an attack, and arguably should not, since the finite-key bound uses the same reading; but it means that the monitor needs its own health check, for example a redundant detector on the monitor tap or a periodic recalibration against a reference, before DCS alarms can be read as

attacks. Second, the residuals must not be sensitive to slow basis-dependent efficiencies, which is why the basis-balance residuals are left out. A mismatch between the two detectors of one basis, which the model omits, would enter the error residuals at first order and would have to be tracked by the refit. Third, the sequential detectors were calibrated with afterpulses present; without them their false-alarm rate at $d = 4$ rose three- to thirtyfold. Fourth, the DCS needs per-window error counts in the key basis, or else the weaker DCS-X.

The DCS was robust to the other operational events we tried: a fivefold step in dark counts, a 20% drop in detector efficiency, a 7 dB loss step, a laser re-lock that moves all intensities by 3% with the monitor following, a better misalignment compensator and a fivefold drop in afterpulse probability (Table 7). At $\alpha = 10^{-3}$ a single link raises about 8.4 DCS alarm episodes per day, and the time to the first false alarm has a median of 2.5 hours at $d = 1$; for a network the alarm budget scales with the number of links, so the operating point has to be chosen per network, as an in-control run length rather than a per-window rate.

8.5 Threats to validity

Internal. All results come from one simulator. The device model (2) with afterpulses, per-basis misalignment, basis-dependent efficiencies and intensity drift is richer than the standard one, but it omits detector dead time, efficiency mismatch between the two detectors of a basis, time-shift side channels and correlations between consecutive pulses. Each attack is a model of the attack, not an implementation, and all are pulse-wise independent; the decoy bounds of [18] with Hoeffding deviations are rigorous only in that setting, and an attacker who changes strategy in response to alarms is not modelled. Seven of the ten faked-state attackers know the current channel parameters; Blind-static, Blind-track and Blind-lag show what less knowledge costs. The breach test (7) is one-sided and counts only bits Eve knows exactly; for the faked-state attacks it follows from their construction. The finite-key analysis substitutes block-mean monitored intensities into bounds that assume known intensities, a heuristic that the simulation checks against the true single-photon counts but that rigorous treatments [26, 28, 29] would replace.

External. The drift amplitudes are assumptions guided by qualitative reports [10, 11], not fitted to a link. The scale d and the sensitivity runs bracket them but cannot replace field data. Every conclusion about false-alarm rates is conditional on this drift model, and recorded telemetry from an operating link is the necessary next test. The DCS needs 10-second windows of public data and costs microseconds per window, so its decision latency is set by the window and by the classical channel, not by computation. The protocol is three-intensity decoy-state BB84 with one finite-key analysis; other protocols and tighter analyses [27] would change key rates but not the indistinguishability argument.

Construct. The ML baselines follow the feature sets and training styles of the cited work but are not reimplementations of those papers; in particular we did not reproduce the adversarial training of [17]. The default calibration of each detector is a design choice, and Table 9 reports the other choice. Per-window rates at one α are one operating point; the appendix gives three. The CUSUM and Block statistics are serially dependent, so their per-window rates are not directly comparable with those of the single-window detectors.

Statistical. Ten seeds give bootstrap intervals that are wide for rare events: a false-alarm rate of 10^{-3} over 8 640 windows per seed rests on a handful of alarms per seed. The intervals in Table 3 reflect this. For Blind-cal the AUC interval excludes 0.5 in 3 of 65 detector–drift cells, about what 95% intervals give under exact equality.

9 Conclusion

Count telemetry cannot certify the absence of a detector-control attack. A faked-state attacker who reproduces the honest photon-number yields and error probabilities leaves the joint law of all window counts unchanged, so every count-telemetry detector works at chance while the finite-key analysis outputs a key the attacker knows almost entirely. In our simulations the hypothesis was met closely enough by an attacker with an ordinary threshold detector, which only a post-hoc block-level statistic separated, and by an attacker who estimated the channel from five minutes of public counts, which no detector separated at any operating point. Against attacks inside the device model the decoy-state bound already protects the key, and detection serves operations rather than secrecy.

Between those cases lies the useful work of a monitor: catching attackers whose emulation is imperfect without raising alarms on a healthy, drifting link. In this simulation model, detectors that compare absolute levels with drift-free calibration data alarmed in most benign windows under nominal drift, and drift-calibrated learned detectors did not hold their target in either direction. Detectors built on a per-window refit of the honest channel held a false-alarm rate near 10^{-3} up to fourfold drift with drift-free calibration only, provided an intensity monitor is present (and, for the sequential forms, unbiased) and the residuals avoid basis-dependent efficiencies. They caught naive blinding at once and in-model attacks reliably; their sequential forms caught part of the error-free and background-free emulation; and none caught emulation whose parameters stay physically plausible. The simulations also showed that slow intensity drift invalidates the single-photon bound with nominal intensities, so per-window intensity monitoring is needed both for the key and for the monitor. The next steps are tests on recorded telemetry from an operating link, adaptive attackers, and monitors that combine count telemetry with timing and coincidence statistics. Code and per-seed results are released with this article.

Data availability statement

The simulator, detectors, finite-key code, per-seed results and the scripts that produce every table and figure are openly available under the MIT licence at `https://github.com/osmancancet/qkd-telemetry-ids`.

A Supplementary tables

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Table 2: Attacker configurations. "In model" marks attacks covered by the decoy-state security proof; the others violate its detector assumptions. Faked-state variants share the mechanism described in the text.

Name	Mechanism	In model
IR, f	Intercept-resend in a random basis on a fraction f of non-empty pulses with an ideal detector; one resend yield for all $n \geq 1$, chosen so that Q_{μ_1} is unchanged	yes
PNS	Lossless channel; single photons blocked, one photon of each multi-photon pulse forwarded with the probability that matches Q_{μ_1} (capped at 1); Eve knows η_B and holds the photon in a quantum memory until the basis announcement [24]	yes
Loss $+\delta$	Extra attenuation of δ dB (denial of service)	yes
Blind-naive	Eve detects every non-empty pulse (a unit-efficiency threshold detector in a random basis) and forwards each detection with a fixed probability chosen to match Q_{μ_1}, so y_n is the same for all $n \geq 1$; no background, no deliberate errors	no
Blind-gain	Reproduces every y_n, including injected background clicks, but adds no errors to non-empty pulses; no afterpulses	no
Blind-nobg	Reproduces the photon part of y_n and the misalignment errors; no background clicks (dark counts or afterpulses)	no
Blind-thresh	Eve's own threshold detector behind an attenuator: she forwards each of her clicks and injects a random-bit click with probability $2Y_0$ when her detector is silent, so $y_n = Y_0 + (1/2 - Y_0)[1 - (1 - \eta_E)^n]$ with η_E set to match Q_{μ_1}; misalignment errors and afterpulses emulated	no
Blind-cal	Reproduces $(y_n, e_{b,n})$ of the current honest channel exactly, afterpulses included (Proposition 1)	no
Blind-static	As Blind-cal, for the nominal channel of Table 1 without any drift	no
Blind-track	As Blind-cal, but tracks only the fibre loss; detector efficiency, dark counts, misalignment and afterpulses stay at nominal values	no
Blind-lag	As Blind-cal, with η (detector part), Y_0 and e_d^b estimated from public counts of the 30 windows before onset and then frozen; tracks the fibre loss	no
Blind-sub	As Blind-cal, but with background lowered to $0.3\,Y_0(t)$ and misalignment to $0.6\,e_d^b(t)$, just above the floors of DCS+F (Section 7.5)	no
Blind-partial	Blind-naive on a random 10% of pulses, honest channel on the rest	no

Table 3: Per-window false-alarm rate (%) on 24-hour benign streams at $\alpha = 0.1\%$: mean over 40 seeds with 95% bootstrap interval. †: threshold from drifting benign data ($d \in \{0.5, 1, 2\}$); all others from drift-free data. The QBER rule uses its fixed 11% limit.

Detector	$d = 0$	$d = 0.25$	$d = 1$	$d = 2$	$d = 4$
QBER	0.00 [0.00, 0.00]	0.00 [0.00, 0.00]	0.00 [0.00, 0.00]	0.00 [0.00, 0.00]	0.00 [0.00, 0.00]
Band	0.10 [0.09, 0.11]	89.31 [89.06, 89.54]	97.27 [97.19, 97.34]	98.65 [98.61, 98.69]	99.42 [99.40, 99.44]
DCS	0.11 [0.10, 0.13]	0.11 [0.09, 0.12]	0.10 [0.09, 0.11]	0.10 [0.09, 0.11]	0.10 [0.08, 0.11]
CUSUM	0.15 [0.11, 0.21]	0.11 [0.08, 0.16]	0.11 [0.08, 0.15]	0.11 [0.08, 0.15]	0.17 [0.12, 0.22]
Block	0.09 [0.07, 0.13]	0.12 [0.09, 0.17]	0.10 [0.07, 0.15]	0.08 [0.05, 0.13]	0.12 [0.08, 0.18]
IF-stat	0.10 [0.09, 0.11]	41.52 [39.39, 44.20]	95.05 [94.60, 95.46]	99.13 [99.01, 99.23]	99.88 [99.86, 99.90]
IF-drift†	0.00 [0.00, 0.00]	0.00 [0.00, 0.00]	0.00 [0.00, 0.00]	0.55 [0.39, 0.81]	30.71 [28.42, 33.05]
IF-res	0.11 [0.10, 0.13]	0.10 [0.09, 0.11]	0.10 [0.08, 0.11]	0.09 [0.08, 0.11]	0.08 [0.07, 0.09]
RF-stat	0.09 [0.08, 0.11]	6.34 [4.76, 8.60]	74.57 [72.36, 76.65]	92.92 [92.16, 93.61]	98.86 [98.66, 99.03]
RF-DR†	0.00 [0.00, 0.00]	0.00 [0.00, 0.00]	0.01 [0.00, 0.05]	0.64 [0.46, 0.90]	8.43 [5.27, 13.28]
RF-res†	0.11 [0.09, 0.12]	0.11 [0.10, 0.13]	0.11 [0.09, 0.12]	0.09 [0.08, 0.11]	0.08 [0.07, 0.09]
DCS-X	0.10 [0.09, 0.11]	0.10 [0.09, 0.12]	0.11 [0.09, 0.12]	0.10 [0.09, 0.12]	0.12 [0.11, 0.13]
Band-roll†	0.00 [0.00, 0.01]	0.00 [0.00, 0.01]	0.00 [0.00, 0.01]	0.69 [0.43, 1.31]	30.20 [29.05, 31.55]

Table 4: TPR at the default threshold, $\alpha = 0.1\%$, $d = 1$ (mean over 40 seeds). The first row is the benign false-alarm rate of the same detector; a TPR close to it is not detection. *: in the training distribution; †: drift-calibrated threshold.

Attack	QBER	Band	DCS	CUSUM	Block	IF-stat	IF-drift†	IF-res	RF-stat	RF-DR†	RF-res†	DCS-X	Band-roll†
benign FPR	0.000	0.973	0.001	0.001	0.001	0.951	0.000	0.001	0.746	0.000	0.001	0.001	0.000
IR, $f = 0.05$	0.000	1.000	0.731	0.999	0.991	0.999	0.001	0.363	1.000*	0.133*	0.928*	0.175	0.047
IR, $f = 0.2$	0.000	1.000	1.000	1.000	0.998	1.000	0.564	0.768	1.000	0.815	1.000	1.000	0.252
PNS	0.000	1.000	1.000	1.000	1.000	1.000	0.143	1.000	1.000*	0.942*	1.000*	1.000	0.251
Blind-naive	0.000	1.000	1.000	1.000	1.000	1.000	0.426	0.998	1.000*	0.999*	1.000*	1.000	0.257
Blind-gain	0.000	1.000	0.029	0.941	0.821	1.000	0.115	0.014	1.000	0.926	0.006	0.004	0.250
Blind-nobg	0.000	1.000	0.008	0.343	0.894	1.000	0.090	0.004	1.000	0.480	0.000	0.015	0.250
Blind-thresh	0.000	0.979	0.001	0.004	0.480	0.956	0.000	0.001	0.733	0.000	0.002	0.002	0.000
Blind-cal	0.000	0.972	0.001	0.001	0.001	0.954	0.000	0.001	0.747	0.000	0.001	0.001	0.000
Blind-static	0.000	0.001	0.001	0.001	0.001	0.007	0.000	0.001	0.005	0.000	0.001	0.001	0.000
Blind-track	0.000	0.001	0.001	0.001	0.002	0.282	0.000	0.001	0.385	0.000	0.001	0.001	0.000
Blind-lag	0.000	0.998	0.001	0.002	0.001	0.949	0.000	0.001	0.770	0.000	0.001	0.001	0.000
Blind-sub	0.000	1.000	0.001	0.000	0.002	1.000	0.028	0.001	0.998	0.081	0.001	0.001	0.176
Blind-partial	0.000	0.997	0.949	1.000	0.992	0.901	0.000	0.114	0.673	0.000	0.507	0.948	0.174
Loss +1 dB	0.000	0.975	0.001	0.000	0.000	0.998	0.000	0.001	0.972	0.001	0.000	0.001	0.000
Loss +3 dB	0.000	0.978	0.001	0.000	0.000	0.999	0.013	0.001	1.000	0.061	0.000	0.000	0.000

Table 5: Finite-key audit at $d = 1$ over all complete blocks inside the attack on a fixed grid (442 blocks per scenario over 40 seeds, 5 760 for the honest streams). K_E/n_Z is the ratio of summed known bits to summed sifted key bits; "Breach" is the share of blocks satisfying (7).

Scenario	$P(\ell > 0)$	Rate ($\times 10^{-4}$)	K_E/n_Z	Breach	Breach $\mid \ell > 0$
Honest (24 h)	1.000	7.66	0.000	0.000	0.000
IR, $f = 0.05$	1.000	3.87	0.025	0.000	0.000
IR, $f = 0.2$	0.000	0.00	0.100	0.000	–
PNS	0.000	0.00	0.990	0.000	–
Blind-naive	1.000	23.24	1.000	1.000	1.000
Blind-gain	1.000	17.23	1.000	1.000	1.000
Blind-nobg	1.000	8.78	1.000	1.000	1.000
Blind-thresh	1.000	7.54	0.990	1.000	1.000
Blind-cal	1.000	7.69	0.990	1.000	1.000
Blind-static	1.000	9.37	0.990	1.000	1.000
Blind-track	1.000	9.35	0.990	1.000	1.000
Blind-lag	1.000	7.75	0.993	1.000	1.000
Blind-sub	1.000	10.90	0.997	1.000	1.000
Blind-partial	1.000	8.78	0.100	0.000	0.000
Loss +1 dB	1.000	5.82	0.000	0.000	0.000
Loss +3 dB	1.000	3.09	0.000	0.000	0.000

Table 6: Breached blocks over 40 seeds at $d = 1$, how many of them each detector would still have released under the block-wise policy (a block with an alarm in any of its windows is withheld; fewer is better), and the share of the paired honest blocks that the same policy withholds (the cost of false alarms).

		Breached blocks released					Honest blocks withheld (%)				
Attack	Breached	DCS	CUSUM	Block	DCS-X	RF-DR	DCS	CUSUM	Block	DCS-X	RF-DR
Blind-naive	442	0	0	0	0	0	6	2	1	7	0
Blind-gain	441	73	0	4	342	4	5	1	1	5	0
Blind-nobg	440	280	138	3	179	206	6	2	1	5	0
Blind-thresh	441	407	429	77	391	439	6	2	1	6	0
Blind-cal	440	408	430	434	406	439	4	1	1	6	0
Blind-static	440	416	427	438	416	440	7	1	2	9	0
Blind-track	440	411	433	434	412	440	7	1	0	5	0
Blind-lag	440	406	430	436	409	440	5	1	1	3	0
Blind-sub	440	414	438	437	418	374	6	1	1	4	0

Table 7: Per-window false-alarm rate (%) under benign operational events outside the drift model, applied to the second half of a 24-hour stream at $d = 1$ (mean over ten fresh seeds), and the AUC of Blind-sub, which keeps its parameters just above the DCS+F floors.

Event (second half of a 24-h stream, $d = 1$)	DCS	CUSUM	Block	DCS-X	DCS+F
Afterpulse probability $\times 0.2$	0.11	0.02	0.10	0.12	48.36
Better compensator ($e_d \times 0.45$)	0.13	0.20	0.16	0.11	19.55
Loss step +7 dB	0.07	0.05	0.08	0.08	47.68
Dark counts $\times 5$	0.11	0.06	0.18	0.10	0.11
Detector efficiency -20%	0.09	0.06	0.16	0.09	0.09
Laser re-lock, intensities +3% (monitored)	0.12	0.07	0.13	0.12	0.12
Decoy monitor calibration step +1%	6.53	49.98	49.98	7.33	6.53
Blind-sub, AUC	0.50	0.49	0.49	0.51	0.50

Table 8: One-factor sensitivity (ten seeds per row, 24-hour streams; nominal setting as in Table 1: 10 s windows, 50 km, $\sigma_I = 1\%$, $p_{\mathrm{ap}} = 1\%$, $(e^Z_{\mathrm{mis}}, e^X_{\mathrm{mis}}) = (1.0\%, 1.2\%)$, intensity drift 1%, $\sigma_M = 0.2\%$, no monitor bias, path efficiency 0.5%, PDL 0.25%, misalignment growth 1%/h). FPR at $\alpha = 0.1\%$; TPR at $d = 1$; "Viol. nom.": honest blocks at $d = 1$ (of 1440) in which $s_{Z,1}$ with nominal intensities exceeded the true count.

	FPR (%), $d = 1$				FPR (%), $d = 4$				DCS TPR, $d = 1$				Block TPR	Viol.
Setting	DCS	CUSUM	Block	DCS-7	DCS	CUSUM	Block	DCS-X	IR	Naive	Gain	Nobg	Thresh	nom.
Nominal	0.08	0.04	0.08	39.40	0.08	0.06	0.13	0.12	0.69	1.00	0.03	0.01	0.51	5
Window 1 s	0.09	0.08	0.07	1.64	0.08	0.07	0.11	0.08	0.02	1.00	0.06	0.00	0.91	0
Window 60 s	0.12	0.01	0.06	81.85	0.11	0.06	0.22	0.17	1.00	1.00	0.08	0.06	0.05	1
σ_I 0.5%	0.09	0.18	0.05	38.44	0.11	0.20	0.08	0.13	0.72	1.00	0.03	0.01	0.56	4
σ_I 2%	0.08	0.06	0.07	38.18	0.08	0.24	0.11	0.11	0.73	1.00	0.02	0.01	0.49	7
Fibre 25 km	0.09	0.09	0.13	69.00	0.12	0.29	0.38	0.13	1.00	1.00	0.13	0.03	0.98	3
Fibre 75 km	0.08	0.09	0.12	11.63	0.10	0.06	0.36	0.10	0.16	1.00	0.01	0.00	0.01	0
p_{ap} 0%	0.09	0.10	0.16	37.97	0.17	1.65	0.45	0.13	0.97	1.00	0.02	0.00	0.53	0
p_{ap} 5%	0.10	0.06	0.09	37.40	0.09	0.16	0.10	0.10	0.22	1.00	0.02	0.01	0.51	13
$(e^Z_{\mathrm{mis}}, e^X_{\mathrm{mis}})$ (0.01,0.01)	0.11	0.09	0.12	36.19	0.09	0.16	0.08	0.12	0.73	1.00	0.03	0.01	0.53	1
$(e^Z_{\mathrm{mis}}, e^X_{\mathrm{mis}})$ (0.01,0.02)	0.11	0.15	0.10	38.91	0.10	0.21	0.13	0.12	0.70	1.00	0.03	0.01	0.54	3
Int. drift 0.5%	0.09	0.10	0.15	39.08	0.09	0.11	0.03	0.11	0.74	1.00	0.03	0.01	0.57	0
Int. drift 2%	0.08	0.10	0.10	39.32	0.09	0.13	0.12	0.12	0.74	1.00	0.03	0.01	0.59	113
Int. monitor off	1.11	54.43	59.85	40.24	31.14	99.49	90.42	32.47	0.20	1.00	0.03	0.02	0.67	5
σ_M 0.5%	0.07	0.07	0.06	39.11	0.08	0.15	0.15	0.12	0.39	1.00	0.02	0.01	0.05	5
σ_M 1%	0.08	0.07	0.06	37.62	0.07	0.12	0.13	0.13	0.17	1.00	0.01	0.01	0.00	5
Monitor bias 0.2%	0.11	3.66	0.25	35.88	0.12	7.78	0.19	0.17	0.62	1.00	0.02	0.01	0.00	0
Monitor bias 0.5%	0.13	3.88	0.33	33.82	0.17	13.45	0.09	0.26	0.35	1.00	0.02	0.00	0.00	0
Monitor bias 1%	0.16	3.16	0.26	31.39	0.21	13.53	0.08	0.31	0.10	1.00	0.01	0.00	0.00	0
Path eff. 0%	0.09	0.11	0.23	6.80	0.11	0.12	0.20	0.11	0.72	1.00	0.02	0.01	0.51	5
Path eff. 1%	0.09	0.12	0.10	66.58	0.10	0.15	0.24	0.09	0.67	1.00	0.03	0.01	0.62	5
Path eff. 2%	0.10	0.15	0.10	83.14	0.10	0.14	0.19	0.12	0.71	1.00	0.03	0.01	0.56	5
PDL 0%	0.09	0.08	0.06	30.14	0.10	0.12	0.10	0.10	0.69	1.00	0.03	0.01	0.61	5
PDL 0.5%	0.07	0.18	0.07	57.18	0.08	0.06	0.18	0.10	0.67	1.00	0.03	0.01	0.56	5
Mis. rate 0.25%/h	0.09	0.07	0.12	39.48	0.11	0.12	0.04	0.10	0.76	1.00	0.03	0.01	0.53	5

Table 9: Per-window false-alarm rate (%) with the other calibration set, the opposite of each detector's default: drift-free data for IF-drift, RF-DR and RF-res, drifting data for all others. Mean with 95% BCa interval over 40 seeds.

Detector	$d = 0$	$d = 0.25$	$d = 1$	$d = 2$	$d = 4$
QBER	0.00 [0.00, 0.00]	0.00 [0.00, 0.00]	0.00 [0.00, 0.00]	0.00 [0.00, 0.00]	0.00 [0.00, 0.00]
Band	0.00 [0.00, 0.00]	0.00 [0.00, 0.00]	0.00 [0.00, 0.00]	0.66 [0.39, 1.26]	27.07 [25.88, 28.43]
DCS	0.12 [0.10, 0.13]	0.11 [0.10, 0.12]	0.10 [0.09, 0.11]	0.10 [0.09, 0.11]	0.09 [0.08, 0.11]
CUSUM	0.16 [0.12, 0.22]	0.13 [0.10, 0.19]	0.11 [0.08, 0.15]	0.11 [0.09, 0.15]	0.17 [0.13, 0.24]
Block	0.13 [0.09, 0.19]	0.15 [0.10, 0.23]	0.13 [0.09, 0.18]	0.10 [0.07, 0.15]	0.14 [0.10, 0.20]
IF-stat	0.00 [0.00, 0.00]	0.00 [0.00, 0.00]	0.04 [0.03, 0.05]	0.30 [0.26, 0.35]	1.28 [1.15, 1.43]
IF-drift†	0.11 [0.09, 0.12]	0.02 [0.02, 0.03]	1.34 [1.01, 2.28]	42.79 [40.65, 45.54]	90.24 [89.37, 91.07]
IF-res	0.12 [0.11, 0.13]	0.11 [0.10, 0.12]	0.11 [0.10, 0.12]	0.10 [0.09, 0.11]	0.09 [0.08, 0.10]
RF-stat	0.00 [0.00, 0.00]	0.00 [0.00, 0.00]	0.06 [0.05, 0.06]	0.26 [0.25, 0.29]	0.48 [0.44, 0.54]
RF-DR†	0.11 [0.10, 0.12]	0.09 [0.07, 0.11]	2.78 [2.29, 3.40]	22.76 [19.83, 25.49]	48.56 [42.59, 54.25]
RF-res†	0.09 [0.08, 0.11]	0.10 [0.09, 0.11]	0.10 [0.08, 0.11]	0.08 [0.07, 0.09]	0.07 [0.06, 0.08]
DCS-X	0.10 [0.09, 0.11]	0.11 [0.10, 0.12]	0.11 [0.10, 0.13]	0.10 [0.09, 0.11]	0.12 [0.11, 0.13]
Band-roll†	0.10 [0.09, 0.11]	0.11 [0.10, 0.12]	0.10 [0.09, 0.12]	8.08 [7.25, 8.81]	44.04 [42.99, 45.04]

Table 10: Per-window false-alarm rate (%) at $\alpha = 1\%$, default calibration. Mean with 95% BCa interval over 40 seeds.

Detector	$d = 0$	$d = 0.25$	$d = 1$	$d = 2$	$d = 4$
QBER	0.00 [0.00, 0.00]	0.00 [0.00, 0.00]	0.00 [0.00, 0.00]	0.00 [0.00, 0.00]	0.00 [0.00, 0.00]
Band	1.01 [0.97, 1.05]	91.39 [91.16, 91.59]	97.82 [97.75, 97.88]	98.95 [98.92, 98.99]	99.61 [99.59, 99.63]
DCS	1.02 [0.98, 1.05]	0.99 [0.95, 1.03]	0.98 [0.95, 1.02]	0.97 [0.93, 1.00]	0.93 [0.89, 0.97]
CUSUM	1.10 [0.97, 1.26]	1.05 [0.94, 1.18]	0.94 [0.84, 1.05]	0.99 [0.88, 1.11]	0.98 [0.86, 1.10]
Block	0.95 [0.82, 1.10]	0.95 [0.78, 1.13]	1.03 [0.89, 1.21]	0.82 [0.70, 0.96]	0.95 [0.81, 1.09]
IF-stat	1.05 [1.02, 1.08]	67.99 [66.42, 69.75]	98.70 [98.55, 98.82]	99.84 [99.81, 99.87]	99.98 [99.97, 99.99]
IF-drift†	0.00 [0.00, 0.00]	0.00 [0.00, 0.00]	0.00 [0.00, 0.00]	3.54 [2.89, 4.38]	49.73 [47.75, 51.81]
IF-res	1.02 [0.98, 1.06]	1.00 [0.97, 1.04]	0.98 [0.95, 1.01]	0.96 [0.93, 0.99]	0.94 [0.91, 0.97]
RF-stat	0.97 [0.93, 1.01]	26.53 [22.61, 30.43]	90.78 [88.53, 92.55]	97.95 [97.27, 98.47]	99.71 [99.57, 99.79]
RF-DR†	0.00 [0.00, 0.00]	0.00 [0.00, 0.00]	0.20 [0.12, 0.34]	3.79 [3.09, 4.62]	18.70 [14.16, 24.87]
RF-res†	1.07 [1.02, 1.11]	1.06 [1.02, 1.11]	1.03 [0.99, 1.07]	0.99 [0.95, 1.03]	0.91 [0.87, 0.96]
DCS-X	1.01 [0.99, 1.04]	1.00 [0.97, 1.04]	1.01 [0.97, 1.04]	1.01 [0.97, 1.06]	1.07 [1.04, 1.10]
Band-roll†	0.03 [0.02, 0.04]	0.03 [0.02, 0.05]	0.03 [0.02, 0.07]	3.59 [2.89, 4.64]	37.12 [35.81, 38.58]

Table 11: Per-window false-alarm rate (%) at $\alpha = 0.01\%$, default calibration. Mean with 95% BCa interval over 40 seeds.

Detector	$d = 0$	$d = 0.25$	$d = 1$	$d = 2$	$d = 4$
QBER	0.00 [0.00, 0.00]	0.00 [0.00, 0.00]	0.00 [0.00, 0.00]	0.00 [0.00, 0.00]	0.00 [0.00, 0.00]
Band	0.01 [0.01, 0.01]	87.44 [87.11, 87.76]	96.78 [96.69, 96.87]	98.41 [98.36, 98.46]	99.27 [99.25, 99.29]
DCS	0.01 [0.01, 0.01]	0.01 [0.01, 0.01]	0.01 [0.01, 0.01]	0.01 [0.01, 0.02]	0.01 [0.01, 0.01]
CUSUM	0.04 [0.02, 0.07]	0.03 [0.01, 0.05]	0.02 [0.01, 0.06]	0.03 [0.01, 0.05]	0.07 [0.04, 0.11]
Block	0.03 [0.01, 0.05]	0.04 [0.02, 0.07]	0.02 [0.01, 0.05]	0.03 [0.01, 0.07]	0.04 [0.02, 0.07]
IF-stat	0.01 [0.01, 0.02]	20.61 [18.61, 23.22]	86.53 [84.98, 87.88]	96.75 [96.25, 97.15]	99.38 [99.24, 99.47]
IF-drift†	0.00 [0.00, 0.00]	0.00 [0.00, 0.00]	0.00 [0.00, 0.00]	0.15 [0.09, 0.25]	20.16 [17.80, 22.53]
IF-res	0.02 [0.01, 0.02]	0.02 [0.01, 0.02]	0.01 [0.01, 0.02]	0.01 [0.01, 0.02]	0.01 [0.01, 0.02]
RF-stat	0.01 [0.01, 0.01]	1.99 [1.04, 3.51]	62.73 [60.38, 65.06]	87.76 [86.34, 88.94]	97.00 [96.38, 97.47]
RF-DR†	0.00 [0.00, 0.00]	0.00 [0.00, 0.00]	0.00 [0.00, 0.00]	0.17 [0.11, 0.26]	5.09 [2.89, 8.75]
RF-res†	0.01 [0.01, 0.02]	0.01 [0.01, 0.02]	0.01 [0.01, 0.02]	0.01 [0.01, 0.01]	0.01 [0.01, 0.01]
DCS-X	0.01 [0.01, 0.01]	0.02 [0.01, 0.02]	0.01 [0.01, 0.02]	0.02 [0.01, 0.02]	0.02 [0.01, 0.02]
Band-roll†	0.00 [0.00, 0.00]	0.00 [0.00, 0.00]	0.00 [0.00, 0.00]	0.27 [0.12, 0.64]	27.95 [26.80, 29.23]

Table 12: Alarm episodes (maximal runs of consecutive alarmed windows) per 24-hour benign stream at $\alpha = 0.1\%$, default calibration. Mean with 95% BCa interval over 40 seeds.

Detector	$d = 0$	$d = 0.25$	$d = 1$	$d = 2$	$d = 4$
QBER	0.0 [0.0, 0.0]	0.0 [0.0, 0.0]	0.0 [0.0, 0.0]	0.0 [0.0, 0.0]	0.0 [0.0, 0.0]
Band	8.8 [8.1, 9.7]	161.7 [157.4, 166.2]	44.3 [42.9, 45.5]	24.6 [23.8, 25.7]	18.1 [17.4, 18.8]
DCS	9.6 [8.5, 10.9]	9.2 [8.1, 10.7]	8.4 [7.4, 9.6]	8.4 [7.5, 9.6]	8.2 [7.2, 9.3]
CUSUM	3.8 [2.9, 4.9]	3.1 [2.5, 3.9]	3.5 [2.6, 4.4]	3.1 [2.4, 4.0]	3.8 [3.0, 4.8]
Block	2.1 [1.6, 2.6]	3.3 [2.4, 4.3]	3.2 [2.3, 4.2]	1.7 [1.2, 2.2]	2.8 [2.0, 3.8]
IF-stat	8.8 [8.0, 9.7]	1288.1 [1257.8, 1309.7]	230.2 [211.9, 250.3]	49.7 [44.6, 56.3]	8.4 [7.3, 9.6]
IF-drift†	0.0 [0.0, 0.0]	0.0 [0.0, 0.0]	0.0 [0.0, 0.0]	21.1 [15.4, 29.2]	325.1 [308.5, 344.2]
IF-res	9.7 [8.7, 11.1]	8.5 [7.6, 9.6]	8.2 [7.3, 9.3]	8.0 [7.0, 9.1]	6.8 [5.9, 7.9]
RF-stat	8.1 [7.0, 9.4]	348.0 [284.6, 423.1]	472.1 [446.4, 501.8]	148.3 [137.9, 160.2]	36.4 [31.7, 42.6]
RF-DR†	0.0 [0.0, 0.0]	0.0 [0.0, 0.0]	0.7 [0.3, 1.6]	22.4 [16.2, 30.4]	77.0 [56.8, 102.5]
RF-res†	9.2 [8.1, 10.4]	9.8 [8.7, 11.1]	9.2 [8.2, 10.5]	8.1 [7.2, 9.2]	7.1 [6.3, 8.2]
DCS-X	8.4 [7.6, 9.4]	9.0 [7.8, 10.1]	9.1 [8.1, 10.3]	8.7 [7.5, 9.9]	10.2 [9.1, 11.3]
Band-roll†	0.2 [0.1, 0.5]	0.2 [0.1, 0.5]	0.4 [0.1, 1.2]	5.3 [4.0, 7.4]	26.9 [25.6, 28.5]

Table 13: Paired ROC AUC at $d = 0$: mean over 40 seeds, with the half-width of the 95% BCa interval as subscript. *: in the training distribution of the supervised detector; ‡: interpolated between training strengths.

Attack	QBER	Band	DCS	CUSUM	Block	IF-stat	IF-drift†	IF-res	RF-stat	RF-DR†	RF-res†	DCS-X	Band-roll†
IR, $f = 0.05$	$1.00_{.00}$	$1.00_{.00}$	$1.00_{.00}$	$1.00_{.00}$	$1.00_{.00}$	$1.00_{.00}$	$0.91_{.01}$	$0.99_{.00}$	$1.00_{.00}{}^*$	$0.90_{.06}{}^*$	$1.00_{.00}{}^*$	$0.91_{.00}$	$0.64_{.00}$
IR, $f = 0.2^\ddagger$	$1.00_{.00}$	$1.00_{.00}$	$1.00_{.00}$	$1.00_{.00}$	$1.00_{.00}$	$1.00_{.00}$	$1.00_{.00}$	$1.00_{.00}$	$1.00_{.00}$	$1.00_{.00}$	$1.00_{.00}$	$1.00_{.00}$	$0.66_{.00}$
PNS	$0.07_{.00}$	$1.00_{.00}$	$1.00_{.00}$	$1.00_{.00}$	$1.00_{.00}$	$1.00_{.00}$	$1.00_{.00}$	$1.00_{.00}$	$1.00_{.00}{}^*$	$1.00_{.00}{}^*$	$1.00_{.00}{}^*$	$1.00_{.00}$	$0.58_{.00}$
Blind-naive	$0.00_{.00}$	$1.00_{.00}$	$1.00_{.00}$	$1.00_{.00}$	$1.00_{.00}$	$1.00_{.00}$	$1.00_{.00}$	$1.00_{.00}$	$1.00_{.00}{}^*$	$1.00_{.00}{}^*$	$1.00_{.00}{}^*$	$1.00_{.00}$	$0.71_{.00}$
Blind-gain	$0.00_{.00}$	$1.00_{.00}$	$0.63_{.00}$	$0.99_{.00}$	$0.98_{.01}$	$1.00_{.00}$	$1.00_{.00}$	$0.63_{.01}$	$1.00_{.00}$	$1.00_{.00}$	$0.52_{.00}$	$0.48_{.01}$	$0.64_{.00}$
Blind-nobg	$0.00_{.00}$	$1.00_{.00}$	$0.59_{.00}$	$0.91_{.01}$	$0.99_{.00}$	$1.00_{.00}$	$1.00_{.00}$	$0.59_{.00}$	$1.00_{.00}$	$1.00_{.00}$	$0.50_{.00}$	$0.37_{.00}$	$0.64_{.00}$
Blind-thresh	$0.49_{.00}$	$0.54_{.00}$	$0.53_{.00}$	$0.57_{.01}$	$0.97_{.01}$	$0.50_{.00}$	$0.50_{.01}$	$0.53_{.00}$	$0.50_{.00}$	$0.50_{.00}$	$0.53_{.01}$	$0.53_{.00}$	$0.51_{.00}$
Blind-cal	$0.50_{.00}$	$0.50_{.00}$	$0.50_{.01}$	$0.50_{.01}$	$0.51_{.03}$	$0.50_{.00}$	$0.50_{.00}$	$0.50_{.01}$	$0.50_{.00}$	$0.50_{.00}$	$0.50_{.00}$	$0.50_{.00}$	$0.50_{.00}$
Blind-static	$0.50_{.00}$	$0.50_{.00}$	$0.50_{.01}$	$0.50_{.01}$	$0.51_{.02}$	$0.50_{.00}$	$0.50_{.00}$	$0.50_{.01}$	$0.50_{.00}$	$0.50_{.00}$	$0.50_{.00}$	$0.50_{.00}$	$0.50_{.00}$
Blind-track	$0.50_{.01}$	$0.50_{.00}$	$0.50_{.00}$	$0.50_{.01}$	$0.51_{.02}$	$0.50_{.00}$	$0.50_{.00}$	$0.50_{.00}$	$0.50_{.00}$	$0.50_{.00}$	$0.50_{.00}$	$0.50_{.00}$	$0.50_{.00}$
Blind-lag	$0.50_{.02}$	$0.53_{.00}$	$0.50_{.01}$	$0.50_{.01}$	$0.52_{.03}$	$0.51_{.00}$	$0.50_{.01}$	$0.50_{.01}$	$0.50_{.00}$	$0.50_{.00}$	$0.50_{.00}$	$0.50_{.00}$	$0.50_{.00}$
Blind-sub	$0.00_{.00}$	$1.00_{.00}$	$0.50_{.01}$	$0.50_{.01}$	$0.48_{.03}$	$1.00_{.00}$	$1.00_{.00}$	$0.50_{.01}$	$1.00_{.00}$	$0.73_{.07}$	$0.50_{.00}$	$0.50_{.00}$	$0.64_{.00}$
Blind-partial	$0.00_{.00}$	$1.00_{.00}$	$1.00_{.00}$	$1.00_{.00}$	$1.00_{.00}$	$0.99_{.00}$	$0.88_{.01}$	$0.97_{.00}$	$0.93_{.04}$	$0.56_{.06}$	$1.00_{.00}$	$1.00_{.00}$	$0.65_{.00}$
Loss +1 dB	$0.61_{.01}$	$0.56_{.00}$	$0.49_{.01}$	$0.48_{.01}$	$0.49_{.03}$	$1.00_{.00}$	$1.00_{.00}$	$0.49_{.01}$	$1.00_{.00}$	$0.54_{.05}$	$0.50_{.00}$	$0.49_{.01}$	$0.52_{.00}$
Loss +3 dB	$0.83_{.00}$	$0.76_{.00}$	$0.47_{.00}$	$0.45_{.01}$	$0.48_{.02}$	$1.00_{.00}$	$1.00_{.00}$	$0.47_{.00}$	$1.00_{.00}$	$0.94_{.06}$	$0.49_{.00}$	$0.47_{.00}$	$0.55_{.00}$

Table 14: Paired ROC AUC at $d = 0.25$: mean over 40 seeds, with the half-width of the 95% BCa interval as subscript. *: in the training distribution of the supervised detector; ‡: interpolated between training strengths.

Attack	QBER	Band	DCS	CUSUM	Block	IF-stat	IF-drift†	IF-res	RF-stat	RF-DR†	RF-res†	DCS-X	Band-roll†
IR, $f = 0.05$	$1.00_{.00}$	$1.00_{.00}$	$1.00_{.00}$	$1.00_{.00}$	$1.00_{.00}$	$0.93_{.01}$	$0.99_{.00}$	$0.99_{.00}$	$1.00_{.00}{}^*$	$0.92_{.05}{}^*$	$1.00_{.00}{}^*$	$0.91_{.00}$	$0.64_{.00}$
IR, $f = 0.2^\ddagger$	$1.00_{.00}$	$1.00_{.00}$	$1.00_{.00}$	$1.00_{.00}$	$1.00_{.00}$	$0.97_{.02}$	$1.00_{.00}$	$1.00_{.00}$	$1.00_{.00}$	$1.00_{.00}$	$1.00_{.00}$	$1.00_{.00}$	$0.66_{.00}$
PNS	$0.45_{.00}$	$1.00_{.00}$	$1.00_{.00}$	$1.00_{.00}$	$1.00_{.00}$	$0.99_{.00}$	$1.00_{.00}$	$1.00_{.00}$	$1.00_{.00}{}^*$	$1.00_{.00}{}^*$	$1.00_{.00}{}^*$	$1.00_{.00}$	$0.58_{.00}$
Blind-naive	$0.00_{.00}$	$1.00_{.00}$	$1.00_{.00}$	$1.00_{.00}$	$1.00_{.00}$	$1.00_{.00}$	$1.00_{.00}$	$1.00_{.00}$	$1.00_{.00}{}^*$	$1.00_{.00}{}^*$	$1.00_{.00}{}^*$	$1.00_{.00}$	$0.70_{.00}$
Blind-gain	$0.00_{.00}$	$1.00_{.00}$	$0.64_{.00}$	$0.99_{.00}$	$0.99_{.00}$	$1.00_{.00}$	$1.00_{.00}$	$0.64_{.00}$	$1.00_{.00}$	$1.00_{.00}$	$0.52_{.00}$	$0.48_{.00}$	$0.62_{.00}$
Blind-nobg	$0.00_{.00}$	$0.82_{.04}$	$0.60_{.00}$	$0.93_{.02}$	$0.99_{.00}$	$1.00_{.00}$	$1.00_{.00}$	$0.59_{.01}$	$1.00_{.00}$	$1.00_{.00}$	$0.50_{.00}$	$0.38_{.00}$	$0.64_{.00}$
Blind-thresh	$0.50_{.00}$	$0.50_{.00}$	$0.53_{.00}$	$0.56_{.01}$	$0.97_{.01}$	$0.50_{.00}$	$0.50_{.00}$	$0.52_{.01}$	$0.50_{.00}$	$0.50_{.00}$	$0.53_{.00}$	$0.53_{.01}$	$0.51_{.00}$
Blind-cal	$0.50_{.00}$	$0.50_{.00}$	$0.50_{.00}$	$0.49_{.01}$	$0.51_{.02}$	$0.50_{.00}$	$0.50_{.00}$	$0.50_{.00}$	$0.50_{.00}$	$0.50_{.00}$	$0.50_{.00}$	$0.50_{.00}$	$0.50_{.00}$
Blind-static	$0.02_{.01}$	$0.03_{.01}$	$0.50_{.00}$	$0.50_{.01}$	$0.50_{.03}$	$0.04_{.01}$	$0.78_{.01}$	$0.50_{.00}$	$0.33_{.03}$	$0.50_{.00}$	$0.50_{.00}$	$0.50_{.00}$	$0.49_{.01}$
Blind-track	$0.02_{.00}$	$0.02_{.00}$	$0.50_{.00}$	$0.51_{.01}$	$0.50_{.02}$	$0.11_{.01}$	$0.80_{.01}$	$0.50_{.00}$	$0.36_{.03}$	$0.50_{.00}$	$0.50_{.00}$	$0.50_{.00}$	$0.49_{.00}$
Blind-lag	$0.46_{.09}$	$0.46_{.09}$	$0.50_{.00}$	$0.51_{.01}$	$0.52_{.03}$	$0.47_{.06}$	$0.51_{.05}$	$0.50_{.00}$	$0.46_{.05}$	$0.50_{.00}$	$0.50_{.00}$	$0.50_{.01}$	$0.48_{.01}$
Blind-sub	$0.00_{.00}$	$1.00_{.00}$	$0.50_{.01}$	$0.50_{.01}$	$0.53_{.03}$	$1.00_{.00}$	$1.00_{.00}$	$0.50_{.01}$	$1.00_{.00}$	$0.72_{.07}$	$0.50_{.00}$	$0.51_{.01}$	$0.63_{.00}$
Blind-partial	$0.16_{.01}$	$0.29_{.01}$	$1.00_{.00}$	$1.00_{.00}$	$1.00_{.00}$	$0.39_{.05}$	$0.80_{.01}$	$0.97_{.00}$	$0.57_{.05}$	$0.51_{.01}$	$1.00_{.00}$	$1.00_{.00}$	$0.65_{.00}$
Loss +1 dB	$0.51_{.00}$	$0.51_{.00}$	$0.49_{.00}$	$0.48_{.01}$	$0.50_{.03}$	$0.95_{.01}$	$1.00_{.00}$	$0.49_{.00}$	$0.98_{.02}$	$0.52_{.03}$	$0.50_{.00}$	$0.49_{.00}$	$0.51_{.00}$
Loss +3 dB	$0.55_{.00}$	$0.55_{.00}$	$0.47_{.00}$	$0.44_{.01}$	$0.46_{.03}$	$0.97_{.01}$	$1.00_{.00}$	$0.47_{.00}$	$1.00_{.00}$	$0.94_{.06}$	$0.49_{.00}$	$0.46_{.00}$	$0.54_{.00}$

Table 15: Paired ROC AUC at $d = 2$: mean over 40 seeds, with the half-width of the 95% BCa interval as subscript. *: in the training distribution of the supervised detector; ‡: interpolated between training strengths.

Attack	QBER	Band	DCS	CUSUM	Block	IF-stat	IF-drift†	IF-res	RF-stat	RF-DR†	RF-res†	DCS-X	Band-roll†
IR, $f = 0.05$	$0.90_{.01}$	$0.90_{.01}$	$0.99_{.00}$	$1.00_{.00}$	$1.00_{.00}$	$0.70_{.02}$	$0.89_{.01}$	$0.99_{.00}$	$0.83_{.03}{}^*$	$0.87_{.03}{}^*$	$1.00_{.00}{}^*$	$0.91_{.00}$	$0.59_{.02}$
IR, $f = 0.2^\ddagger$	$1.00_{.00}$	$1.00_{.00}$	$1.00_{.00}$	$1.00_{.00}$	$1.00_{.00}$	$0.70_{.03}$	$1.00_{.00}$	$1.00_{.00}$	$0.85_{.03}$	$0.99_{.01}$	$1.00_{.00}$	$1.00_{.00}$	$0.64_{.02}$
PNS	$0.49_{.00}$	$0.76_{.03}$	$1.00_{.00}$	$1.00_{.00}$	$1.00_{.00}$	$0.90_{.02}$	$0.99_{.01}$	$1.00_{.00}$	$0.76_{.03}{}^*$	$1.00_{.00}{}^*$	$1.00_{.00}{}^*$	$1.00_{.00}$	$0.63_{.01}$
Blind-naive	$0.00_{.00}$	$0.68_{.04}$	$1.00_{.00}$	$1.00_{.00}$	$1.00_{.00}$	$0.98_{.03}$	$0.99_{.01}$	$1.00_{.00}$	$0.67_{.07}{}^*$	$1.00_{.00}{}^*$	$1.00_{.00}{}^*$	$1.00_{.00}$	$0.50_{.02}$
Blind-gain	$0.00_{.00}$	$0.75_{.04}$	$0.64_{.01}$	$0.99_{.01}$	$0.98_{.00}$	$0.97_{.04}$	$0.99_{.01}$	$0.64_{.01}$	$0.45_{.05}$	$1.00_{.00}$	$0.52_{.00}$	$0.48_{.00}$	$0.40_{.01}$
Blind-nobg	$0.33_{.01}$	$0.36_{.01}$	$0.59_{.01}$	$0.85_{.02}$	$0.98_{.01}$	$0.92_{.03}$	$0.98_{.01}$	$0.59_{.01}$	$0.39_{.04}$	$0.99_{.02}$	$0.50_{.00}$	$0.39_{.01}$	$0.65_{.01}$
Blind-thresh	$0.50_{.00}$	$0.50_{.00}$	$0.53_{.01}$	$0.58_{.02}$	$0.96_{.01}$	$0.50_{.00}$	$0.50_{.00}$	$0.53_{.01}$	$0.50_{.00}$	$0.50_{.00}$	$0.53_{.01}$	$0.53_{.01}$	$0.50_{.00}$
Blind-cal	$0.50_{.00}$	$0.50_{.00}$	$0.50_{.00}$	$0.50_{.01}$	$0.50_{.03}$	$0.50_{.00}$	$0.50_{.00}$	$0.49_{.00}$	$0.50_{.00}$	$0.50_{.00}$	$0.50_{.00}$	$0.50_{.00}$	$0.50_{.00}$
Blind-static	$0.01_{.01}$	$0.00_{.00}$	$0.50_{.01}$	$0.51_{.01}$	$0.52_{.02}$	$0.00_{.00}$	$0.12_{.03}$	$0.50_{.01}$	$0.01_{.01}$	$0.37_{.02}$	$0.50_{.00}$	$0.50_{.01}$	$0.32_{.02}$
Blind-track	$0.01_{.01}$	$0.00_{.00}$	$0.51_{.00}$	$0.51_{.01}$	$0.51_{.03}$	$0.02_{.01}$	$0.39_{.05}$	$0.51_{.00}$	$0.08_{.01}$	$0.36_{.02}$	$0.50_{.00}$	$0.50_{.00}$	$0.31_{.02}$
Blind-lag	$0.58_{.09}$	$0.58_{.09}$	$0.50_{.00}$	$0.50_{.01}$	$0.50_{.02}$	$0.54_{.06}$	$0.56_{.08}$	$0.50_{.00}$	$0.58_{.08}$	$0.54_{.07}$	$0.50_{.00}$	$0.50_{.01}$	$0.22_{.02}$
Blind-sub	$0.07_{.01}$	$0.18_{.01}$	$0.50_{.00}$	$0.50_{.01}$	$0.51_{.02}$	$0.84_{.06}$	$0.95_{.02}$	$0.50_{.00}$	$0.25_{.03}$	$0.59_{.09}$	$0.50_{.00}$	$0.50_{.00}$	$0.52_{.01}$
Blind-partial	$0.40_{.01}$	$0.41_{.01}$	$1.00_{.00}$	$1.00_{.00}$	$1.00_{.00}$	$0.43_{.01}$	$0.44_{.02}$	$0.97_{.00}$	$0.42_{.01}$	$0.45_{.01}$	$1.00_{.00}$	$1.00_{.00}$	$0.60_{.01}$
Loss +1 dB	$0.50_{.00}$	$0.50_{.00}$	$0.49_{.00}$	$0.48_{.01}$	$0.48_{.03}$	$0.65_{.04}$	$0.80_{.05}$	$0.49_{.00}$	$0.44_{.02}$	$0.50_{.04}$	$0.50_{.00}$	$0.49_{.00}$	$0.51_{.00}$
Loss +3 dB	$0.51_{.00}$	$0.51_{.00}$	$0.47_{.00}$	$0.45_{.01}$	$0.49_{.03}$	$0.71_{.03}$	$0.95_{.02}$	$0.47_{.00}$	$0.50_{.02}$	$0.84_{.05}$	$0.49_{.00}$	$0.47_{.00}$	$0.52_{.00}$

Table 16: Paired ROC AUC at $d = 4$: mean over 40 seeds, with the half-width of the 95% BCa interval as subscript. *: in the training distribution of the supervised detector; ‡: interpolated between training strengths.

Attack	QBER	Band	DCS	CUSUM	Block	IF-stat	IF-drift†	IF-res	RF-stat	RF-DR†	RF-res†	DCS-X	Band-roll†
IR, $f = 0.05$	$0.75_{.02}$	$0.75_{.02}$	$0.98_{.00}$	$1.00_{.00}$	$1.00_{.00}$	$0.59_{.02}$	$0.74_{.01}$	$0.97_{.00}$	$0.67_{.02}{}^*$	$0.76_{.02}{}^*$	$0.99_{.00}{}^*$	$0.90_{.01}$	$0.50_{.02}$
IR, $f = 0.2^\ddagger$	$1.00_{.00}$	$1.00_{.00}$	$1.00_{.00}$	$1.00_{.00}$	$1.00_{.00}$	$0.60_{.02}$	$0.89_{.02}$	$1.00_{.00}$	$0.69_{.03}$	$0.99_{.00}$	$1.00_{.00}$	$1.00_{.00}$	$0.62_{.03}$
PNS	$0.50_{.00}$	$0.57_{.01}$	$1.00_{.00}$	$1.00_{.00}$	$1.00_{.00}$	$0.82_{.03}$	$0.82_{.02}$	$1.00_{.00}$	$0.65_{.04}{}^*$	$0.99_{.01}{}^*$	$1.00_{.00}{}^*$	$1.00_{.00}$	$0.65_{.01}$
Blind-naive	$0.00_{.00}$	$0.37_{.03}$	$1.00_{.00}$	$1.00_{.00}$	$1.00_{.00}$	$0.97_{.05}$	$0.74_{.04}$	$1.00_{.00}$	$0.49_{.07}{}^*$	$1.00_{.01}{}^*$	$1.00_{.00}{}^*$	$1.00_{.00}$	$0.43_{.02}$
Blind-gain	$0.00_{.00}$	$0.37_{.03}$	$0.65_{.01}$	$0.98_{.01}$	$0.97_{.02}$	$0.96_{.03}$	$0.72_{.03}$	$0.65_{.01}$	$0.32_{.06}$	$0.98_{.03}$	$0.53_{.01}$	$0.47_{.01}$	$0.32_{.01}$
Blind-nobg	$0.41_{.01}$	$0.41_{.01}$	$0.58_{.00}$	$0.79_{.03}$	$0.96_{.02}$	$0.93_{.02}$	$0.78_{.02}$	$0.57_{.01}$	$0.24_{.04}$	$0.94_{.03}$	$0.50_{.00}$	$0.40_{.01}$	$0.64_{.01}$
Blind-thresh	$0.50_{.00}$	$0.50_{.00}$	$0.53_{.01}$	$0.58_{.03}$	$0.93_{.03}$	$0.50_{.00}$	$0.50_{.00}$	$0.52_{.01}$	$0.50_{.01}$	$0.50_{.00}$	$0.53_{.01}$	$0.53_{.01}$	$0.50_{.00}$
Blind-cal	$0.50_{.00}$	$0.50_{.00}$	$0.50_{.01}$	$0.50_{.01}$	$0.51_{.02}$	$0.50_{.00}$	$0.50_{.00}$	$0.50_{.01}$	$0.50_{.00}$	$0.50_{.00}$	$0.50_{.00}$	$0.50_{.01}$	$0.50_{.00}$
Blind-static	$0.01_{.01}$	$0.00_{.00}$	$0.51_{.00}$	$0.52_{.01}$	$0.51_{.03}$	$0.00_{.00}$	$0.02_{.01}$	$0.51_{.00}$	$0.01_{.00}$	$0.22_{.03}$	$0.50_{.00}$	$0.50_{.01}$	$0.25_{.01}$
Blind-track	$0.02_{.01}$	$0.00_{.00}$	$0.51_{.00}$	$0.51_{.01}$	$0.49_{.02}$	$0.01_{.00}$	$0.22_{.04}$	$0.51_{.00}$	$0.04_{.01}$	$0.24_{.03}$	$0.50_{.00}$	$0.50_{.01}$	$0.24_{.02}$
Blind-lag	$0.53_{.08}$	$0.53_{.08}$	$0.50_{.00}$	$0.49_{.01}$	$0.49_{.03}$	$0.52_{.06}$	$0.55_{.08}$	$0.50_{.00}$	$0.55_{.07}$	$0.56_{.07}$	$0.50_{.00}$	$0.50_{.00}$	$0.18_{.02}$
Blind-sub	$0.16_{.01}$	$0.19_{.01}$	$0.50_{.00}$	$0.50_{.01}$	$0.49_{.03}$	$0.71_{.06}$	$0.55_{.03}$	$0.50_{.00}$	$0.14_{.02}$	$0.44_{.08}$	$0.50_{.00}$	$0.50_{.00}$	$0.45_{.01}$
Blind-partial	$0.43_{.00}$	$0.43_{.00}$	$1.00_{.00}$	$1.00_{.00}$	$1.00_{.00}$	$0.46_{.01}$	$0.44_{.01}$	$0.98_{.00}$	$0.45_{.00}$	$0.45_{.01}$	$1.00_{.00}$	$1.00_{.00}$	$0.54_{.01}$
Loss +1 dB	$0.50_{.00}$	$0.50_{.00}$	$0.49_{.00}$	$0.47_{.01}$	$0.50_{.03}$	$0.54_{.04}$	$0.55_{.04}$	$0.49_{.00}$	$0.46_{.02}$	$0.52_{.03}$	$0.50_{.00}$	$0.48_{.01}$	$0.50_{.00}$
Loss +3 dB	$0.50_{.00}$	$0.50_{.00}$	$0.47_{.00}$	$0.44_{.01}$	$0.47_{.03}$	$0.67_{.03}$	$0.74_{.02}$	$0.47_{.00}$	$0.46_{.03}$	$0.70_{.06}$	$0.49_{.00}$	$0.46_{.00}$	$0.51_{.00}$

Table 17: TPR at the default threshold, $\alpha = 0.1\%$, $d = 4$, with the benign false-alarm rate in the first row (mean over 40 seeds).

Attack	QBER	Band	DCS	CUSUM	Block	IF-stat	IF-drift†	IF-res	RF-stat	RF-DR†	RF-res†	DCS-X	Band-roll†
benign FPR	0.000	0.994	0.001	0.002	0.001	0.999	0.307	0.001	0.989	0.084	0.001	0.001	0.302
IR, $f = 0.05$	0.000	1.000	0.525	0.998	0.990	1.000	0.619	0.273	1.000*	0.260*	0.780*	0.160	0.283
IR, $f = 0.2$	0.011	1.000	1.000	1.000	0.997	1.000	0.940	0.745	1.000	0.891	1.000	1.000	0.476
PNS	0.000	1.000	1.000	1.000	1.000	1.000	0.750	1.000	1.000*	0.803*	1.000*	1.000	0.470
Blind-naive	0.000	1.000	1.000	1.000	1.000	1.000	0.671	0.997	1.000*	0.986*	1.000*	1.000	0.293
Blind-gain	0.000	1.000	0.040	0.893	0.732	1.000	0.547	0.023	1.000	0.935	0.011	0.005	0.248
Blind-nobg	0.000	1.000	0.007	0.267	0.775	1.000	0.713	0.003	1.000	0.469	0.001	0.015	0.469
Blind-thresh	0.000	0.994	0.002	0.049	0.475	0.999	0.361	0.002	0.984	0.077	0.002	0.002	0.326
Blind-cal	0.000	0.994	0.001	0.001	0.001	0.998	0.314	0.001	0.987	0.101	0.001	0.001	0.329
Blind-static	0.000	0.004	0.001	0.002	0.001	0.294	0.000	0.001	0.224	0.000	0.001	0.001	0.144
Blind-track	0.000	0.004	0.001	0.004	0.003	0.624	0.001	0.001	0.838	0.007	0.001	0.001	0.109
Blind-lag	0.000	1.000	0.001	0.001	0.000	1.000	0.384	0.001	0.999	0.105	0.001	0.001	0.025
Blind-sub	0.000	0.999	0.001	0.001	0.001	1.000	0.243	0.001	0.997	0.090	0.001	0.001	0.256
Blind-partial	0.000	0.996	0.927	0.999	0.993	0.999	0.213	0.147	0.974	0.035	0.488	0.919	0.390
Loss +1 dB	0.000	0.993	0.000	0.000	0.002	0.998	0.373	0.001	0.987	0.115	0.000	0.001	0.349
Loss +3 dB	0.000	0.995	0.001	0.000	0.000	1.000	0.510	0.001	0.999	0.171	0.001	0.001	0.289

Table 18: TPR at matched FPR, $d = 1$: each detector thresholded at the $(1 - \alpha)$ quantile of its own benign scores at the same drift scale, $\alpha = 0.1\%$ (oracle threshold; mean over 40 seeds).

Attack	QBER	Band	DCS	CUSUM	Block	IF-stat	IF-drift†	IF-res	RF-stat	RF-DR†	RF-res†	DCS-X	Band-roll†
IR, $f = 0.05$	0.76	0.76	0.74	1.00	0.99	0.01	0.45	0.38	0.02	0.47	0.92	0.17	0.12
IR, $f = 0.2$	1.00	1.00	1.00	1.00	1.00	0.01	1.00	0.78	0.03	0.97	1.00	1.00	0.26
PNS	0.00	0.05	1.00	1.00	1.00	0.22	1.00	1.00	0.01	1.00	1.00	1.00	0.25
Blind-naive	0.00	0.10	1.00	1.00	1.00	0.76	1.00	1.00	0.02	1.00	1.00	1.00	0.28
Blind-gain	0.00	0.09	0.03	0.94	0.84	0.62	1.00	0.01	0.00	1.00	0.01	0.00	0.25
Blind-nobg	0.00	0.00	0.01	0.35	0.90	0.52	1.00	0.00	0.00	0.95	0.00	0.02	0.25
Blind-thresh	0.00	0.00	0.00	0.00	0.51	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00
Blind-cal	0.00	0.00	0.00	0.00	0.00	0.00	0.01	0.00	0.00	0.00	0.00	0.00	0.00
Blind-static	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00
Blind-track	0.00	0.00	0.00	0.00	0.00	0.00	0.01	0.00	0.00	0.00	0.00	0.00	0.00
Blind-lag	0.00	0.00	0.00	0.00	0.00	0.00	0.01	0.00	0.00	0.00	0.00	0.00	0.00
Blind-sub	0.00	0.00	0.00	0.00	0.00	0.32	1.00	0.00	0.00	0.22	0.00	0.00	0.23
Blind-partial	0.00	0.00	0.95	1.00	0.99	0.00	0.01	0.12	0.00	0.00	0.44	0.95	0.23
Loss +1 dB	0.00	0.00	0.00	0.00	0.00	0.01	0.63	0.00	0.00	0.03	0.00	0.00	0.00
Loss +3 dB	0.00	0.00	0.00	0.00	0.00	0.03	0.98	0.00	0.00	0.26	0.00	0.00	0.01

Table 19: TPR at matched FPR, $d = 4$ (as Table 18).

Attack	QBER	Band	DCS	CUSUM	Block	IF-stat	IF-drift†	IF-res	RF-stat	RF-DR†	RF-res†	DCS-X	Band-roll†
IR, $f = 0.05$	0.04	0.04	0.53	1.00	0.99	0.00	0.00	0.31	0.00	0.03	0.79	0.15	0.02
IR, $f = 0.2$	0.74	0.74	1.00	1.00	1.00	0.00	0.01	0.81	0.00	0.55	1.00	1.00	0.24
PNS	0.00	0.00	1.00	1.00	1.00	0.05	0.03	1.00	0.00	0.31	1.00	1.00	0.25
Blind-naive	0.00	0.00	1.00	1.00	1.00	0.42	0.00	1.00	0.00	0.79	1.00	1.00	0.25
Blind-gain	0.00	0.00	0.04	0.89	0.75	0.34	0.00	0.03	0.00	0.49	0.01	0.01	0.15
Blind-nobg	0.00	0.00	0.01	0.26	0.79	0.27	0.03	0.00	0.00	0.00	0.00	0.01	0.09
Blind-thresh	0.00	0.00	0.00	0.04	0.51	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00
Blind-cal	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00
Blind-static	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.01
Blind-track	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00
Blind-lag	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.01	0.00	0.00	0.00
Blind-sub	0.00	0.00	0.00	0.00	0.00	0.06	0.00	0.00	0.00	0.00	0.00	0.00	0.01
Blind-partial	0.00	0.00	0.93	1.00	0.99	0.00	0.00	0.17	0.00	0.00	0.63	0.91	0.00
Loss +1 dB	0.00	0.00	0.00	0.00	0.00	0.00	0.01	0.00	0.00	0.00	0.00	0.00	0.01
Loss +3 dB	0.00	0.00	0.00	0.00	0.00	0.01	0.03	0.00	0.00	0.00	0.00	0.00	0.00

Table 20: Change in total key length under attack relative to the paired honest copy of the same blocks (%, mean over 40 seeds).

Attack	$d = 0$	$d = 0.25$	$d = 1$	$d = 2$	$d = 4$
IR, $f = 0.05$	-44.3	-45.6	-50.2	-58.7	-65.8
IR, $f = 0.2$	-100.0	-100.0	-100.0	-100.0	-100.0
PNS	-100.0	-100.0	-100.0	-100.0	-100.0
Blind-naive	+145.9	+158.7	+204.3	+298.2	+600.8
Blind-gain	+84.0	+93.2	+127.4	+185.7	+406.7
Blind-nobg	+15.4	+15.7	+17.6	+19.9	+29.9
Blind-thresh	+0.5	+0.5	+0.6	+0.8	+0.9
Blind-cal	+0.1	+0.1	-0.1	-0.1	+0.1
Blind-static	+0.1	+5.2	+24.4	+53.0	+195.2
Blind-track	+0.0	+5.4	+23.3	+56.8	+165.5
Blind-lag	+0.1	+0.2	-0.3	-8.2	-16.1
Blind-sub	+30.1	+32.9	+42.6	+63.6	+124.9
Blind-partial	+9.9	+11.0	+13.9	+20.1	+32.8
Loss +1 dB	-23.9	-24.2	-24.9	-26.4	-28.3
Loss +3 dB	-57.2	-57.7	-59.5	-62.8	-65.8